



Neurons and Synapses

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Like the entomologist hunting for brightly colored butterflies, my attention has chased in the gardens of the grey matter cells with delicate and elegant shapes, the mysterious butterflies of the soul, whose beating of wings may one day reveal to us the secrets of the mind.

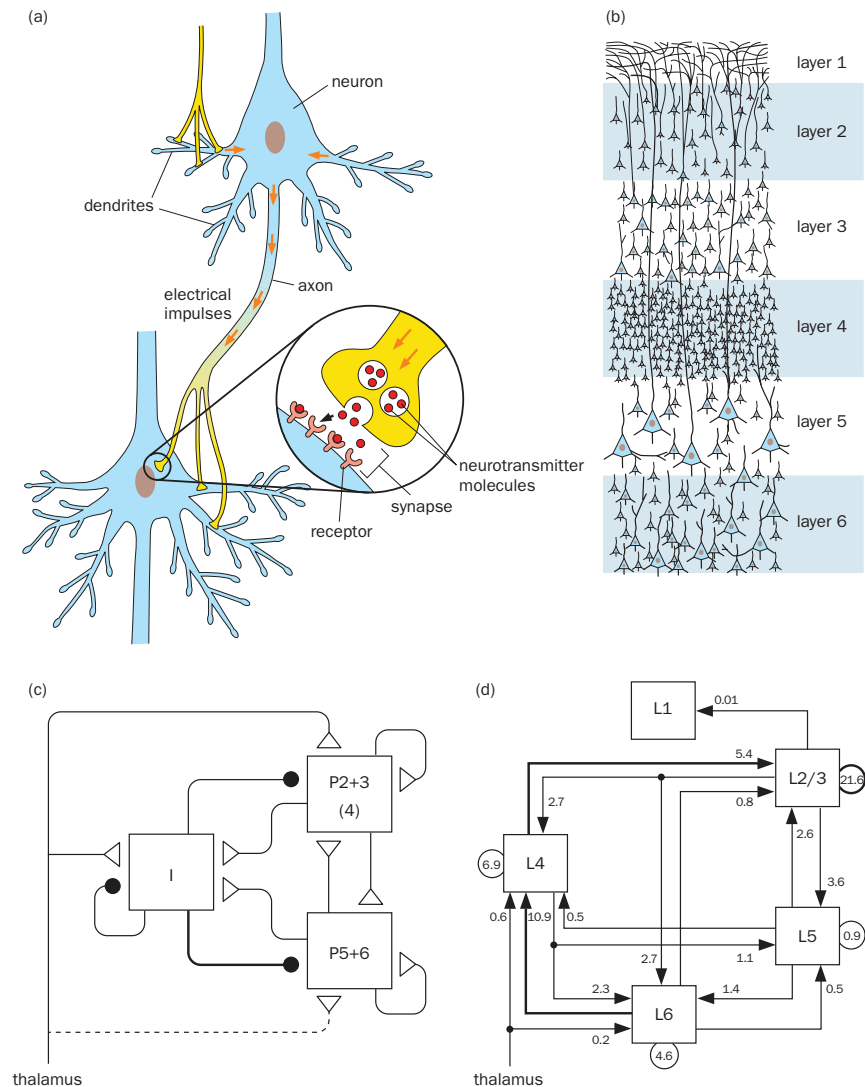
Santiago Ramón y Cajal in *Recollections of My Life*

2.1 INTRODUCTION

A brain is an intricate biological system composed of nerve cells (neurons) as well as non-neuronal glia cells (Cajal 1911). Individual neurons connect each other through junctions called synapses (Sherrington 1906) (**Figure 2.1a**). A neuron receives electrical signals from other neurons at synapses distributed across its dendrite (a branched, tree-like structure). These inputs converge to the soma (the cell body of a neuron), where the inputs are summated. The overall input determines neuronal output signal in the form of stereotypical electrical pulses called action potentials or spikes. Spikes propagate along fibers called axons away from the cell body to synaptic terminals, the presynaptic side of synapses onto postsynaptic neurons. The brain size spans a wide range across species: the worm *C. elegans*, fly *Drosophila* and mouse have 302, 135,000 and 70 million neurons, respectively. The human brain is empowered with 86 billion neurons (Azevedo et al. 2009); the total length of fibers connecting neurons in our head is roughly the distance between the earth and the moon, forming a tremendously complex circuit underlying our mental life.

To face the challenge of understanding the brain dynamics and functions, a good starting point is to consider microcircuits within a brain region dedicated to a specific function such as the retina at the back of the eyeball for vision. Microcircuits come in innumerable varieties; at the same time some general principles of organization are valid across neural microcircuits ranging from sensory organs and motor pattern generators of invertebrates to the primate prefrontal cortex (PFC) often dubbed “the CEO of the brain” in primates (Shepherd 1998; Shepherd and Grillner 2018).

Figure 2.1 Cortical microcircuit. (a) Schematic of two pyramidal neurons, one (upper right) sends an axon to the other (lower left). Insert is the connecting synapse where an action potential in the presynaptic neuron triggers release of neurotransmitters, which travel to the other side of the (20–40 nanometer wide) synaptic cleft and change the activity of the postsynaptic neuron. (b) Neocortex displays six layers across its thickness; only excitatory neurons are shown for the sake of simplicity. (c) Interconnectivity of a layered microcircuit in the neocortex, divided into superficial layers (including also layer 4 spiny stellates) and deep layers. P denotes excitatory pyramidal neurons (also stellate cells in layer 4), I denotes inhibitory interneurons. External inputs to a microcircuit target both superficial excitatory neurons and inhibitory cells. (d) Percentage of all excitatory-to-excitatory synapses for each connection type, indicated by arrows, in the primary visual cortex of the cat. Panel (a) is reproduced from Bear et al. (2015), (b) from Barrett (2017), (c) from Nelson (2002) provided by Kevan Martin, (d) from Binzegger et al. (2004). [Courtesy of Kevan Martin.] [Binzegger, T., R. Douglas, and K. Martin (2004). A quantitative map of the circuit of cat primary visual cortex. *J. Neurosci.* 24(39), 8441–8453. Copyright 2004 Society for Neuroscience.]



The neocortex, a thin (1–3 mm thick) sheet of nerve tissue beneath the pia that covers the brain, is a distinct feature of mammals in the animal kingdom and plays a major role in higher brain functions. In the neocortex, excitatory neurons increase the activity of target neurons, the majority of which are pyramidal cells named for their pyramid-shaped soma (**Figure 2.1a**). The remaining nerve cells are inhibitory, acting on suppressing target neurons. They are often referred to as interneurons because virtually all of them make connections locally within a cortical region. Intriguingly, excitatory:inhibitory cell ratio varies significantly in different cortical regions; in humans, the inhibitory neurons are proportionally more numerous in the PFC (nearly 30% of all neurons) than in the primary visual area V1 (15%) (Krienen et al. 2020). In a given cortical area, primates have higher percentages of interneurons than do rodents; in the motor cortex, the excitatory to inhibitory cell ratio is 5:1 in mouse, 3:1 in marmoset monkey and 2:1 in human (Bakken et al. 2021).

A cortical microcircuit has a laminar structure subdivided into several layers (**Figure 2.1b**); their interconnections are shown in **Figure 2.1c** simplified with “superficial layers 2/3” (including layer 4) and “deep layers 5/6.” It is endowed with local recurrent connection loops; for instance, layer 2/3 pyramidal cells project to neurons in layer 5/6, which in turn project back to layer 2/3. The primary visual cortex (visual area 1 or V1) is the entry point into the neocortex for visual inputs, which arrive in the retina

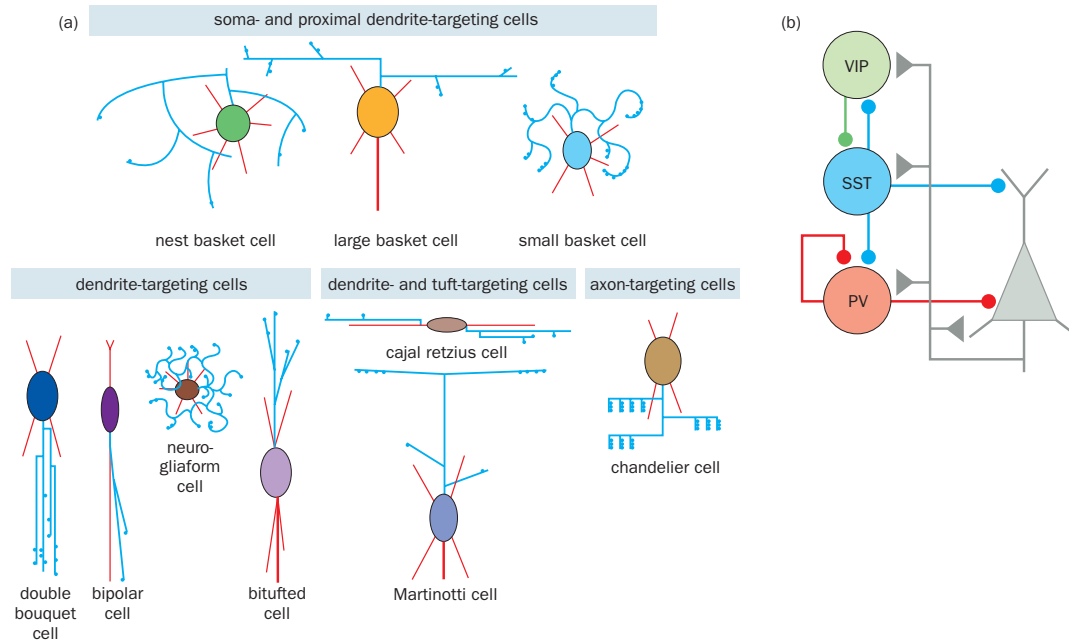


Figure 2.2 Diversity of inhibitory neurons in the neocortex. (a) Scheme summarizing the main anatomical properties of inhibitory interneurons. Each neuron type has a different colored soma; dendrites, red; axons, blue lines; axonal boutons, blue dots. Neurons are orientated with the pia facing upwards. Some interneurons have a prominent vertical dendrite. (b) A disinhibitory motif composed of three types of interneurons, where PV cells target the perisomatic region of a pyramidal cell, and SST cells target the pyramidal dendrite, whereas VIP cells target SST cells. Panel (a) is reproduced from Markram et al. (2004), (b) from Wang and Yang (2018). [Markram, H., Toledo-Rodriguez, M., Wang, Y., Gupta, A., Silberberg, G., & Wu, C. (2004). Interneurons of the neocortical inhibitory system. *Nature Reviews Neuroscience*, 5(10), 793–807.] [Wang, X. J., & Yang, G. R. (2018). A disinhibitory circuit motif and flexible information routing in the brain. *Current Opinion in Neurobiology*, 49, 75–83.]

and are relayed to the cortex through the lateral geniculate nucleus (LGN) in the thalamus. Thalamic inputs constitute about 6% of all excitatory synapses in a microcircuit of V1, although physiologically they powerfully drive the activity of target cells. Recurrent connections between excitatory neurons are especially abundant in the superficial layers 2/3, constituting more than 20% of the bulk of excitatory synapses in a given volume of the cortical tissue (**Figure 2.1d**). This feature holds true for excitatory-to-inhibitory, inhibitory-to-excitatory and inhibitory-to-inhibitory connections (Binzegger et al. 2004). In primates, the intrinsic connections in the superficial layers 2/3 display stripe-like patterns in the PFC in contrast to patch-like patterns in V1 (Levitt et al. 1993; Kritzer and Goldman-Rakic 1995). As we shall see throughout this book, feedback loops within cortical microcircuits play a key role in explaining neural population dynamics and cognitive functions.

Another characteristic of the neocortical microcircuits is a wide diversity of inhibitory cells (**Figure 2.2a**) (Kawaguchi and Kubota 1997; Markram et al. 2004; Paul et al. 2017). Three distinct types can be classified according to their projections and protein expressions (**Figure 2.2b**). The first type targets the peri-somatic region of pyramidal cells and acts to control spiking output of excitatory neurons. Soma-targeting cells (STCs) express the calcium-binding protein parvalbumin (PV) and can be labeled as such. The second type, which either express somatostatin (SST) or the calcium-binding protein calbindin (CB), target pyramidal dendrites. Thus SST/CB interneurons are well positioned to control inputs onto excitatory neurons. The third type, identifiable by expression of either vasoactive intestinal peptide (VIP) or the calcium-binding protein calretinin (CR), preferentially targets SST/CB inhibitory neurons (Freund and Buzsáki 1996; DeFelipe 2002). These three interneuron types form a disinhibitory motif, originally proposed theoretically (Wang et al. 2004), which has been shown experimentally to represent a general feature of cortical microcircuits (**Figure 2.2b**) (Kepecs and Fishell

2014; Tremblay et al. 2016). The fourth major type of inhibitory neurons does not express PV, SST or VIP (Machold et al. 2023): they constitute 90% of interneurons in layer 1 with various firing patterns (Meng et al. 2023); in other layers, they either express neuropeptide Y and mediate GABA_B receptor-dependent slow inhibition (Tamás et al. 2003), or express peptide CCK and underlie the cannabinoid action in the brain (Katona et al. 1999).

We shall discuss microcircuits of interconnected neurons in Chapter 3. Here, we will cover the basics of mathematical models of single neurons and synapses, with reference to basic experimental observations of excitatory pyramidal neurons and inhibitory cells in the mammalian cortex. We will primarily describe electrical signaling, but it is important to keep in mind that neuronal operations are shaped by chemical substances. There are a plethora of chemical processes inside a nerve cell that interact with electrical signaling and may underlie interesting computations. As a simple example of this kind, we will describe single-neuron adaptation mediated by intracellular calcium ions. At a synapse, communication is mediated by the release of chemicals (neurotransmitters) from a presynaptic neuron, and binding of transmitters with a receptor in the postsynaptic neuron. Signaling processes of transmitters and neuromodulators are targets of many substances that affect the brain, including therapeutic medicine for mental illness and addictive drugs. Furthermore, the hallmark of the brain as a biological organ is its ability to adapt from experience and acquire knowledge. In Chapter 4 we shall discuss biological mechanisms of plasticity of synaptic connections believed to underlie learning.

As we shall see, biophysical properties of single neurons and synapses matter a lot to brain dynamics and functions, and they display variations across species (Elston et al. 2001; Gidon et al. 2020; Beaulieu-Laroche et al. 2021). Accurate mathematical description of single neurons affects greatly our understanding of collective network behaviors (Chapter 3); neuronal adaptation plays a key role in decision computations (Chapter 6). Biologically realistic modeling of synaptic transmission is crucial for the demonstration of “cognitive-type” neural circuits capable of both working memory and decision making (Chapters 5 and 6), as well as shedding insights into biological deficits associated with psychiatric disorders (Chapter 10). Models of the four major cell types (excitatory pyramidal cells and three inhibitory subclasses) will be used to realize a gating mechanism for a working memory system to filter out behaviorally irrelevant distracting stimuli (Chapter 5) and more generally in a multi-regional large-scale neocortex (Chapter 9). Therefore, what is discussed in this chapter forms a foundation for neural circuit models of cognitive functions covered in Parts II and III.

2.2 INTEGRATE-AND-FIRE NEURON

The electrical signaling of a neuron is determined by voltage difference across its membrane, $V = V_{in} - V_{out}$, where V_{in} and V_{out} are voltages inside and outside the neuron, respectively. The levels of V_{in} and V_{out} are determined by concentrations of ions such as Cl^- , Na^+ , K^+ and Ca^{2+} , in the cytosol and in the extracellular medium (Hille 2001). A nerve cell has an elaborated spatial structure (Figures 2.1a and 2.2), and the membrane potential varies from a spatial location to another. In general, $V(x, t)$ depends on both the location x of a spatially extended neuron and time t .

The choice of a single neuron model with a certain degree of complexity should be done judiciously depending on the question the model is designed to investigate. A morphologically fully reconstructed pyramidal neuron (Figure 2.3a) may be modeled with hundreds of compartments, each assumed to be isopotential (i.e., the transmembrane potential

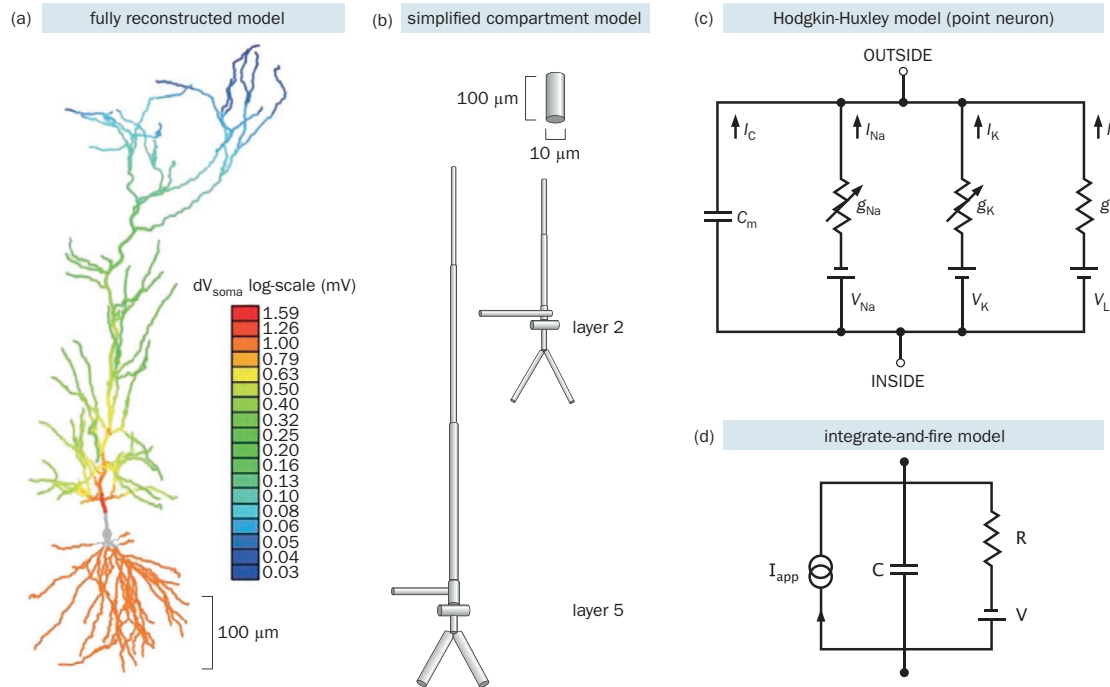


Figure 2.3 Single neuron models. (a) A pyramidal neuron model with a fully reconstructed dendritic tree. (b) A simplified multi-compartmental model of a pyramidal neuron in layer 5 (left) and another one in layer 2/3 (right). (c) A point Hodgkin-Huxley model of action potential. (d) A leaky integrate-and-fire neuron. Panel (a) is reproduced from Nicholson et al. (2006), (b) from Bush and Sejnowski (1993), (c) from Hodgkin and Huxley (1952) and (d) from Koch (1999). [Nicholson, D. A., Trana, R., Katz, Y., Kath, W. L., Spruston, N., & Geinisman, Y. (2006). Distance-dependent differences in synapse number and AMPA receptor expression in hippocampal CA1 pyramidal neurons. *Neuron*, 50(3), 431–442.] [Bush, P. C., & Sejnowski, T. J. (1993). Reduced compartmental models of neocortical pyramidal cells. *Journal of Neuroscience Methods*, 46(2), 159–166.]

everywhere within a compartment is roughly the same) (Rall 1964; Rall and Shepherd 1968; Segev et al. 1995), which is appropriate to study how synaptic inputs propagate and are spatiotemporally integrated throughout the dendrite tree (Spruston 2008; Stuart et al. 2016; Goetz et al. 2021). In neural circuit modeling, such a model is often simplified, reduced approximately to a small number of compartments (**Figure 2.3b**), which captures the basic spatial structure of a neuron with apical and basal dendrites (Bush and Sejnowski 1993), down to just three (Wang et al. 2004) or two (perisomatic and dendritic) compartments (Pinsky and Rinzel 1994; Wang 1999a; Kep-ecs et al. 2002). When distinct dendritic versus somatic signaling is not the main focus, it is reasonable to use a “point neuron” with a single compartment, which can be the Hodgkin-Huxley conductance-based model (**Figure 2.3c**) or the integrate-and-fire model (**Figure 2.3d**). Similarly, modeling of a synapse varies in terms of the degree of complexity. It is important to keep in mind that, on the one hand, it is not the case that the more details a model incorporates, the better; on the other hand, certain model properties of single neurons or synapses may have drastic implications for collective dynamical behavior of a network comprising thousands of neurons.

2.2.1 Neuronal Membrane as an RC Circuit

A neuronal membrane is mostly a fatty lipid bilayer, which is not permeable to charged molecules, and hence acts as a capacitor with a capacitance C by separating the charges inside and outside the neuron. Charge stored in a capacitor is $Q = CV$ and its time derivative defines a capacitive current $dQ / dt = C dV / dt$. There are various types of transmembrane ion channels, which, when opened, selectively allow particular types of ions to

pass through them. For instance, a potassium channel highly prefers K^+ to Na^+ even though Na^+ has the same charge and is smaller than K^+ .

Opening an ion channel leads to a current I that, according to Ohm's law, can be described by $I = g(V - V_{rev})$ (with the unit of $\mu A / cm^2$), where g is a conductance (the inverse of resistance) per unit area (in $\mu S / cm^2$) and V_{rev} is a reversal potential. Given a membrane surface area A , we can write $G = gA$ (in μS) and the total current $I = G(V - V_{rev})$ (in nA) (with a slight abuse of notation for I). The conductance G quantifies the strength of an ion current, but the size of a current is also proportional to its "driving force," $V - V_{rev}$, the difference between the membrane potential V (which varies dynamically over time) and the reversal potential V_{rev} . The value of V_{rev} depends on the selective permeability of an ion channel and the intracellular and extracellular concentrations of ions that can pass through the channel. When a channel is permeable for a single type of ions, V_{rev} is given by the Nernst equation, $V_{rev} = \left(\frac{RT}{Fz}\right) \ln([C_{out}]/[C_{in}])$, where R is the

universal gas constant, T is the temperature, F is the Faraday constant, z is the charge of an ion, and C_{in} and C_{out} are the intracellular and extracellular ionic concentrations. Typically, the potassium concentration is higher inside than outside a neuron, thus V_K is negative; the opposite holds for the sodium and calcium concentrations, thus V_{Na} and V_{Ca} are positive. At room temperature, the value of reversal potential is about -80 mV for a potassium current, $+50$ mV for a sodium current, and $+120$ mV for a calcium current. The combined effects of various ion types yield a net reversal potential given by the Goldman-Hodgkin-Katz equation (Hille 2001).

When a neuron is "at rest," the voltage is lower inside compared with outside the membrane, and V is typically about -70 mV. This is conveniently modeled by a "leak" current $I_L = G_L(V - V_L)$ with a constant G_L and $V_L = -70$ mV. A minimum model for a passive neuronal membrane is described by an RC circuit, with the input resistance $R = 1/G_L$ and the capacitance C associated with the membrane's lipid bilayer. If I_{app} represents an external current injected into the membrane, then according to Kirchhoff's circuit law, the sum of the resistive and capacitive currents must equal to I_{app} , $C_m dV / dt + I_L = I_{app}$, or

$$C_m \frac{dV}{dt} = -G_L(V - V_L) + I_{app} \quad (2.1)$$

where I_{app} is an external stimulus as an injected current. Eq. (2.1) can be rewritten as

$$\frac{dV}{dt} = \frac{(V_{ss} - V)}{\tau_m} \quad (2.2)$$

where $V_{ss} = V_L + I_{app} / G_L$ is the steady-state and $\tau_m = C_m / G_L = RC_m$ is the membrane time constant. For instance, if $C_m = 0.2$ nF, $G_L = 0.02$ μS , then $\tau_m = 10$ ms.

Eq. (2.2) is a differential equation that, unlike an algebraic equation, depends on both a variable V and its time derivative. It is also called a dynamical system, which describes and predicts how the behavior unfolds in time: in a given "state" (a particular value of V), the right-hand side determines the slope (dV / dt) of its change into the future. A "steady-state" V_{ss} is determined by the value of V when it does not change in time, $dV / dt = 0$ yielding $V = V_{ss}$. In the absence of input ($I_{app} = 0$), the steady-state membrane potential is V_L . When driven by an input current I_{app} , V_{ss} increases linearly with I_{app} , and is inversely proportional to the leak conductance G_L (a given amount of excitation produces more depolarization if leak of current across the membrane is smaller). This steady-state is stable: if V is transiently depolarized or hyperpolarized by a current pulse, after the input offset it will evolve back to V_{ss} exponentially with a time constant τ_m . This is a simple example of an attractor state, which is a

mathematical term to describe a state of a dynamical system that is stable under perturbations. More specifically, since the membrane is governed by the linear RC circuit equation, it can be readily solved. For an initial condition $V(t=0) = V_0$, the general solution of Eq. (2.1) is

$$V(t) = V_{ss} + (V_0 - V_{ss})\exp(-t / \tau_m). \quad (2.3)$$

Therefore, as shown in **Figure 2.4b** (black curve), when a current pulse is applied, V evolves over time so that its difference with V_{ss} decreases exponentially. The time it takes for V to reach the steady-state within 1% is $\sim 5 \tau_m$ (at $t = 5\tau_m$, $\exp(-t / \tau_m) = \exp(-5) \simeq 0.7\%$). When the current pulse I_{app} is turned off, it takes a similar amount of time for V to decay back to V_L . For instance, if $\tau_m = 10$ ms, the system “forgets” a previously shown input within about 50 ms.

Neurons are endowed with many active channels, and their conductances are not fixed but depend on voltage or/and intracellular ion concentrations (Linás 1988). Therefore, generally the Kirchhoff’s circuit equation has the form of $C_m dV / dt = -\sum_j I_{ion,j} + I_{app}$, where the sum is over all intrinsic ion currents $I_{ion,j} = g_j (V - V_{rev,j})$. The value of $V_{rev,j}$ relative to V determines whether a current is positive or negative. By convention, at resting membrane $V \simeq -70$ mV, a negative current is called an “inward” current that tends to increase or depolarize the transmembrane potential V , whereas a positive current is called an outward current that tends to decrease or hyperpolarize V . This convention may be a bit confusing at first sight, but it makes sense when we notice the negative sign in front of an ion current in the membrane equation.

2.2.2 LIF as a Simple Spiking Neuron Model

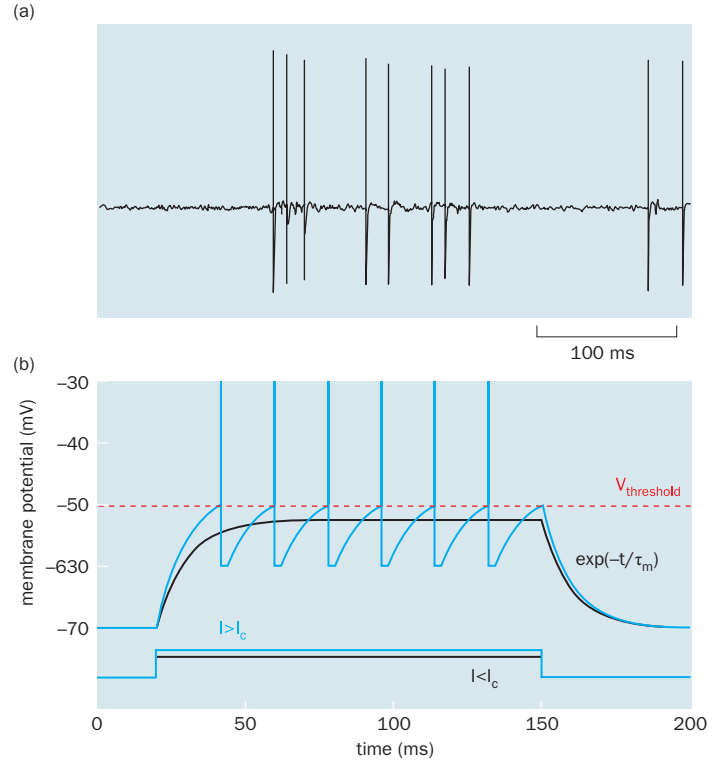
Neuronal communication is conveyed by action potentials, or spikes, which are brief pulses ($\sim 1 - 2$ millisecond duration) of electrical signals (**Figure 2.4a**). The more spikes are emitted per unit time (second), the more active is a neuron. The leaky integrate-and-fire (LIF) model, originally proposed in 1907 by the French physiologist Louis Lapicque (see Brunel and van Rossum 2007), is built on the simplification that spikes are stereotypical and brief events. It describes a passive neuron that incorporates spikes into an RC circuit in a simplified way. Namely, the model assumes the generation of a spike as a “point event” in time whenever V exceeds a preset voltage threshold V_{th} . Specifically, if V reaches V_{th} at time t , a spike is discharged with the spike time t ; V is instantaneously reset to V_{reset} and stays there for a refractory period of time τ_{ref} . The values of V_{th} , V_{reset} and τ_{ref} are parameters that can be determined by physiological measurements for a particular type of neurons that one wishes to model. For the sake of concreteness, let us set as an example $V_{th} = -50$ mV, $V_{reset} = -60$ mV and $\tau_{ref} = 2$ ms.

The behavior of the LIF model neuron depends on the intensity of input current I_{app} . With a small I_{app} , $V(t)$ settles in a steady-state V_{ss} below V_{th} (**Figure 2.4b**, black curve). On the other hand, if I_{app} exceeds a threshold value I_c , the steady-state V_{ss} is larger than V_{th} . The value of I_c is given by imposing $V_{ss} = V_{th}$, or $I_c = G_L (V_{th} - V_L) = 0.4$ nA. For $I_{app} > I_c$, V_{ss} is larger than V_{th} . As $V(t)$ evolves towards V_{ss} , a spike is produced when the membrane potential crosses V_{th} and is reset to V_{reset} for a refractory period τ_{ref} . Afterwards, V rises and over time reaches V_{th} again, leading to repetitive firing of spikes (**Figure 2.4b**, blue curve). With a fixed I_{app} the firing is regular in time, characterized by a constant period T or inter-spike interval (ISI), and firing rate $r = 1 / T$.

The ISI can be obtained analytically from Eq. (2.3). Immediately after a refractory period, $V = V_{reset}$. Using this value as an initial condition, $V(t)$

Figure 2.4 Leaky integrate-and-fire (LIF) model of a spiking neuron.

(a) A sample spike train recorded from a visual cortical area of a behaving monkey. (b) Behavior of the LIF neuron model. In response to an input current I is below a threshold level I_c , the membrane potential reaches a steady-state (black). With an increased I above I_c (blue), the neuron model displays repetitive firing of spikes, which are not explicitly modeled in the model and assumed to be a point process. The horizontal dashed red line indicates the voltage threshold V_{th} for spike firing. Panel (a) is reproduced from Amarasingham et al. (2006). [Amarasingham A, Chen TL, Geman S, Harrison MT, Sheinberg DL. (2006). Spike count reliability and the Poisson hypothesis. *J Neurosci.* 26(3):801–9. Copyright 2006 Society for Neuroscience.]



evolves in time according to Eq. (2.3) and reaches V_{th} at $t = T - \tau_{ref}$. This condition yields an algebraic equation for T . Therefore, one obtains an analytical expression for the firing rate r as a function of the current I_{app} ,

$$r = \begin{cases} \frac{1}{\tau_{ref} + \frac{C_m}{G_L} \ln \left[1 + \frac{G_L \Delta V}{(I_{app} - I_c)} \right]} & \text{if } I_{app} > I_c \\ 0 & \text{otherwise} \end{cases} \quad (2.4)$$

where $\Delta V = (V_{th} - V_{reset})$ is the voltage difference between the firing threshold and reset.

The frequency-current relationship ($f-I$ curve), also called the neuronal input-output transfer function, is a fundamental concept that describes a neuron's mapping from an input current to an output firing rate. For the LIF model, the firing rate starts at zero at $I_{app} = I_c$ and increases steeply near I_c (in fact, the slope of $r(I)$ is infinity at $I_{app} = I_c$). Well above I_c , one can show from Eq. (2.4) that r is approximately linear with I_{app} , with a slope given by $1 / C_m \Delta V$. At very large I_{app} , r cannot exceed $1 / \tau_{ref}$ because ISI must be longer than the refractory period; therefore, the firing rate eventually plateaus. With $\tau_{ref} = 2$ ms, the maximum firing rate is $1 / \tau_{ref} = 500$ Hz.

2.2.3 Spiking Variability

In an intact brain *in vivo*, synaptic inputs to a neuron fluctuate considerably in time (Destexhe et al. 2003). Consequently, neuronal firing of a spike train is highly irregular (see Figure 2.4a). For a periodic spike train, the ISI is constant. By contrast, ISI is a random variable for a stochastic point process. For instance, in a Poisson process, at any small time window dt , an event (a spike) occurs with a probability λdt , and there is no temporal correlation between events. In this case, consecutive $ISIs$

are independent from each other, with an exponential probability density $p(ISI) = \lambda \exp(-\lambda ISI)$, and the mean firing rate is $r = \lambda = 1 / \langle ISI \rangle$.

The stochasticity of a spike train can be quantified in two ways. The first is the coefficient of variation (CV) of ISIs, defined as

$$CV = \frac{\text{Standard Deviation}(ISI)}{\text{Mean}(ISI)} = \frac{\langle (ISI - \langle ISI \rangle)^2 \rangle^{1/2}}{\langle ISI \rangle} \quad (2.5)$$

where $\langle x \rangle = \int p(x) x dx$ denotes the average of x . The second measure is Fano factor of spike count $N(T)$ (number of spikes in a time window T) defined by

$$F(T) = \frac{\text{Variance}(N(T))}{\text{Mean}(N(T))} \quad (2.6)$$

For a regular periodic spike train, both CV and $F(T)$ are 0. At the other extreme, if a spike train is Poisson, we have CV and $F(T)$ equal to 1. High variability represents a salient characteristic of spike trains observed in cortical neurons *in vivo*, for which typically the spike-count variance exceeds the mean (Tolhurst et al. 1983; Softky and Koch 1993; Goris et al. 2014). An example is shown in **Figure 2.5a** for neurons in the primary visual cortex V1 of a macaque monkey in response to a visual stimulus, with a time window of $T = 1$ s, in which case the spike count is the same as firing rate per second. As it can be seen, the variance is larger than the mean, and the Fano factor is larger than 1. This excessive variability can be explained by a statistical model where spike trains is approximated by a Poisson process with the rate λ that itself varies stochastically over time (Goris et al. 2014). Similarly, the CV is typically larger than 1 for neural activity in the prefrontal cortex of a behaving macaque monkey (**Figure 2.5b**) (Compte et al. 2003).

To capture stochastic spike firing in an integrate-and-fire model, one can introduce a noise $\eta(t)$ into I_{app} , so Eq. (2.1) becomes

$$C_m \frac{dV}{dt} = -G_L (V - V_L) + \mu + \sigma \eta(t) \quad (2.7)$$

where μ is the mean of I_{app} , σ is the noise level. The simplest kind of noise has no temporal correlation and a Gaussian probability density for η of zero mean and unit variance, $p(\eta) = 1 / \sqrt{2\pi} \exp(-\eta^2 / 2)$. This is called “white noise,” because its power spectrum is constant and therefore completely devoid of dependence on frequencies. The absence of temporal correlation is mathematically described by $\langle \eta(t) \eta(t') \rangle = \delta(t - t')$, where the delta function $\delta(t)$ is infinity with $t = 0$ and zero otherwise (its integral is 1 and physical unit is one over time). Such a stochastic linear differential equation with a leak and white noise is called the Ornstein-Uhlenbeck process (Wax 1954).

The input current now is noisy with a mean of μ and a standard deviation of σ . When μ is below I_c , the deterministic steady-state V_{ss} is below

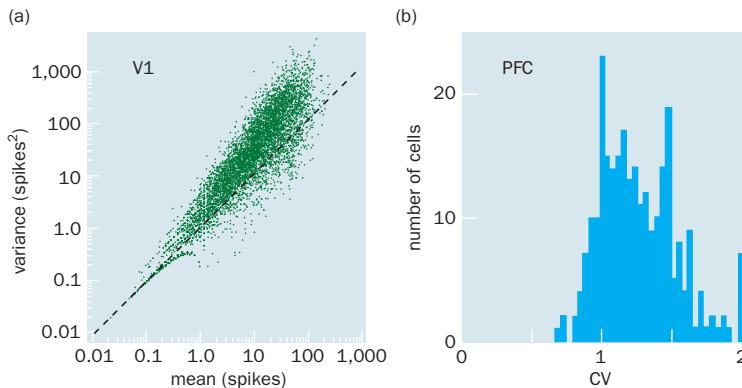


Figure 2.5 Variability of the spiking activity of cortical neurons.

(a) Variance versus mean of spike count in a 1-second time window of neurons recorded from the primary visual cortex V1 of monkey (each point corresponds to one neuron under one stimulus condition). The slope is Fano factor, which is about 1 at low firing rates (around the diagonal line) and larger than 1 at high firing rates (above the diagonal line). (b) The histogram of the coefficient of variation (CV) of interspike intervals across a population of neurons recorded in monkey prefrontal cortex, showing that CV values are mostly close to or larger than 1. (a) is adapted from Goris et al. (2014), (b) from (Compte et al. 2003). [Goris, R. L., Movshon, J. A., & Simoncelli, E. P. (2014). Partitioning neuronal variability. *Nature Neuroscience*, 17(6), 858–865.] [Compte, A., Constantinidis, C., Tegnér, J., Raghavachari, S., Chafee, M. V., Goldman-Rakic, P. S., & Wang, X. J. (2003). Temporally irregular mnemonic persistent activity in prefrontal neurons of monkeys during a delayed response task. *Journal Neurophysiology*, 90(5), 3441–3454.]

the firing threshold, but spiking is now possible because V is driven by noise and may still reach V_{th} through fluctuations (**Figure 2.6a**). In the presence of noise, the interspike interval ISI is a random variable, following a probabilistic distribution. ISIs are statistically independent of each other, since each time a spike is fired, V is reset to V_{reset} and memory about the past history is lost.

Starting at V_{reset} , the time it takes for V to reach V_{th} for the first time is mathematically called “first passage time” (Ricciardi 1977; Tuckwell 1988); an ISI is the first passage time plus the refractory period. Therefore, the statistics of ISIs can be analyzed by the theory of first passage times of an Ornstein-Uhlenbeck process. In particular, the average $\langle ISI \rangle$ is the inverse of the mean firing rate r , and the input-output function is given by (Ricciardi 1977; Renart et al. 2003):

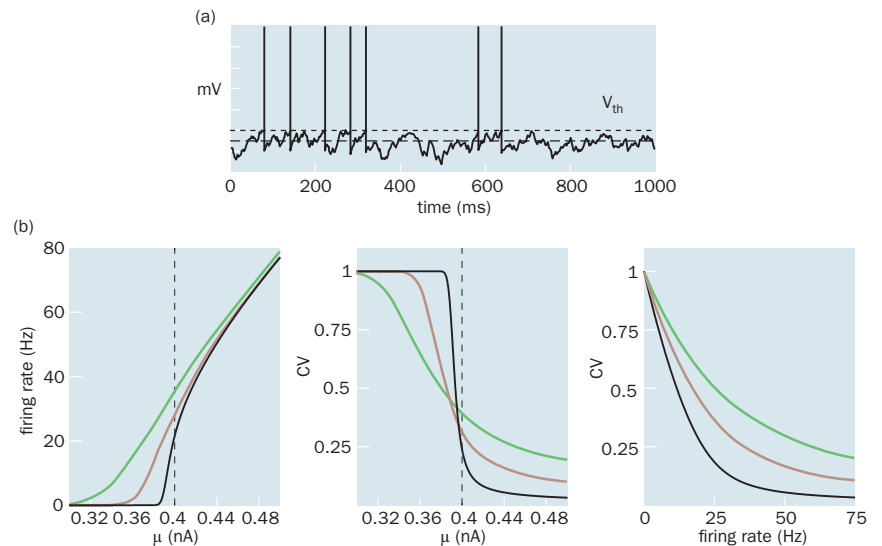
$$\frac{1}{r} = \tau_{ref} + \tau_m \sqrt{\pi} \int \frac{C_m(V_{th}-V_{ss})}{C_m(V_{reset}-V_{ss})} \frac{\sqrt{\tau_m \sigma}}{\sqrt{\tau_m \sigma}} e^{x^2} (1 + \operatorname{erf}(x)) dx \quad (2.8)$$

where $\operatorname{erf}(x) = \left(2/\sqrt{\pi}\right) \int_0^x \exp(-x'^2) dx'$ is the error function. The CV is given by (Renart et al. 2003):

$$CV^2 = 2\pi r^2 \int \frac{C_m(V_{th}-V_{ss})}{C_m(V_{reset}-V_{ss})} e^{x^2} \left[\int_{-\infty}^x e^{y^2} (1 + \operatorname{erf}(y)) dy \right] dx \quad (2.9)$$

Figure 2.6b (left panel) shows the frequency-current relationship as a function of the mean μ , with three σ different values. As it can be seen in **Figure 2.6b** (middle panel), the LIF model driven by noise exhibits two separate regimes as far as variability is concerned. When the mean input is sub-threshold ($\mu < I_c$), spike discharge is triggered by fluctuations in the current, and spike trains are irregular. Therefore, in this regime, the CV is high, close to one. For supra-threshold mean current ($\mu > I_c$), the mean becomes dominant over the noise, and firing becomes increasingly regular. This happens when $(\mu - I_c)$ is much larger than $\sigma_I = \sigma / \sqrt{\tau_m}$ (note that σ has the unit of $\text{nA} \sqrt{\text{ms}}$, thus σ_I has the unit of nA). With large μ , CV decays to zero and spiking becomes regular. The sharpness of this transition depends on σ . When the fluctuations are small, the transition is very

Figure 2.6 LIF model in the presence of noise. (a) Activity driven by a noisy input current with a mean μ near the threshold I_c . (b) The frequency-current curve (left) and CV (middle) as a function of the mean input current μ for three different levels of noise. CV decreases with the mean firing rate (right). Reproduced from Renart et al. (2003). Copyright 2003. From Mean-field theory of recurrent cortical networks: Working memory circuits with irregularly spiking neurons. In J. Feng (Ed.), Computational Neuroscience: A Comprehensive Approach by Renart, A., N. Brunel, and X.-J. Wang. Reproduced by permission of Taylor & Francis Group, LLC, a division of Informa plc.



sharp, so as soon as the neuron starts firing, it does so in a regular fashion. For large values of σ the transition is smooth, so the neuron fires irregularly for μ near I_c and becomes progressively more regular with μ much larger than I_c . When the firing rate increases as a result of an increase in the mean current, CV decreases. This decrease is faster with smaller noise (**Figure 2.6b** (right panel)).

Evidently, a model with a fixed noise level cannot explain the commonly observed high CV or Fano factor of cortical neurons in a broad range of firing rates (**Figure 2.5**). Neuronal variability is important to understand, because it plays a crucial role in neural computation. For instance, cognitive processes such as decision making are probabilistic. To what extent does the source of stochasticity originate from the irregular firing of neurons? A more basic physiological question is how a high variability of neural spike firing can be explained mechanistically. This phenomenon suggests that the total input to a single cortical cell must operate near the firing threshold, regardless of the firing rate level. How can this be realized in a cortical circuit? We will come back to this issue when we discuss the excitation-inhibition balance in recurrent neural networks in Chapter 3.

2.3 CONDUCTANCE-BASED MODELS OF SINGLE NEURONS

2.3.1 Hodgkin-Huxley Formalism of Action Potential

The biophysical mechanism for the generation of an action potential is well understood, starting with the seminal work of Hodgkin and Huxley (1952). The description of this model can be found in standard textbooks (Hille 2001; Nichols et al. 1992). Here we will present it succinctly. According to the Hodgkin-Huxley model, an action potential is produced by an interplay of voltage-gated Na^+ and K^+ ion currents (I_{Na} and I_K). I_{Na} is an inward current and depolarizes the neuronal membrane; by contrast, I_K is an outward current hyperpolarizes the membrane. As usual, we can write $I_{Na} = g_{Na}(V - V_{Na})$ and $I_K = g_K(V - V_K)$ with the reversal potentials V_{Na} and V_K . Unlike a passive leak conductance, the conductances g_{Na} and g_K are not constant but “gated” by membrane potential V . Specifically, the potassium conductance g_K becomes $g_K n^4$, where n is the activation gating variable. The idea is that n represents the fraction of “gates” in an open state, and $1 - n$ is the fraction of gates in a closed state. The gating variable n obeys a dynamical equation according to a first-order chemical kinetics:

$$dn / dt = \alpha_n(1 - n) - \beta_n n \quad (2.10)$$

with the opening and closing rates α_n and β_n dependent on V . This equation can be rewritten as $dn / dt = (n_\infty - n) / \tau_n$, with the steady-state $n_\infty = \alpha_n / (\alpha_n + \beta_n)$ and time constant $\tau_n = 1 / (\alpha_n + \beta_n)$.

The sodium conductance g_{Na} has not only an activation gating variable m but also an inactivation gating variable h , such that g_{Na} becomes $g_{Na} m^3 h$, with m and h described by similar first-order kinetic equations as for n but with different opening and closing rates.

Thus, the Hodgkin-Huxley model consists of four differential equations:

$$C_m \frac{dV}{dt} = -g_L(V - V_L) - g_{Na} m^3 h (V - V_{Na}) - g_K n^4 (V - V_K) + I_{app} \quad (2.11)$$

$$\frac{dm}{dt} = \phi_m(\alpha_m(V)(1 - m) - \beta_m(V)m) = \frac{(m_\infty(V) - m)}{\tau_m(V)} \quad (2.12)$$

$$\frac{dh}{dt} = \phi_h (\alpha_h(V)(1-h) - \beta_h(V)h) = \frac{(h_\infty(V) - h)}{\tau_h(V)} \quad (2.13)$$

$$\frac{dn}{dt} = \phi_n (\alpha_n(V)(1-n) - \beta_n(V)n) = \frac{(n_\infty(V) - n)}{\tau_n(V)} \quad (2.14)$$

where I_{app} is the injected current (in $\mu\text{A}/\text{cm}^2$). The parameters ϕ_x are the temperature factors, equal to 1 in the original Hodgkin-Huxley model (with the kinetic rates measured at 6–7°C). At a higher room or body temperature, chemical reactions are faster, and ϕ_m , ϕ_h and ϕ_n are larger than 1. Note that our convention is to use lowercase g for specific conductance (per cm^2) and uppercase G for total conductance (as in the above LIF model).

The Hodgkin-Huxley model is a highly nonlinear coupled system, as the V equation depends on all four variables, and the rates α and β in the equations for the gating variables are complex functions of V . However, using an experimental technique called “voltage-clamp,” one can hold V at a constant level in time, in which case α and β are mere numbers. By measuring the time course of an ion current such as I_K or I_{Na} and its conductance (equal to $I/(V - V_{Na})$ and $I/(V - V_K)$ with a fixed V , respectively), one can deduce the α and β values for each V . Using this method, Hodgkin and Huxley deduced the quantitative dependence of the rates on V . **Figure 2.7** shows the resulting steady-states and time constants for the three gating variables as a function of V . Explicitly, the rate functions are:

$$\alpha_m(V) = -0.1(V + 40) / (\exp(-0.1(V + 40)) - 1); \quad (2.15)$$

$$\beta_m(V) = 4 \exp(-(V + 65)/18) \quad (2.16)$$

$$\alpha_h(V) = 0.07 \exp(-0.05(V + 65)); \quad (2.17)$$

$$\beta_h(V) = 1 / (1 + \exp(-0.1(V + 35))) \quad (2.18)$$

$$\alpha_n(V) = -0.01(V + 55) / (\exp(-0.1(V + 55)) - 1); \quad (2.19)$$

$$\beta_n(V) = 0.125 \exp(-(V + 65)/80) \quad (2.20)$$

The other parameter values of the classic Hodgkin-Huxley model are: $c_m = 1 \mu\text{F}/\text{cm}^2$; the maximum conductances $g_{Na} = 120$, $g_K = 36$, $g_L = 0.3$ (in mS/cm^2); and the reversal potentials $V_{Na} = +50$, $V_K = -77$, $V_L = -54.4$ (in mV). The passive time constant $\tau_m = c_m / g_L \simeq 3.3$ ms.

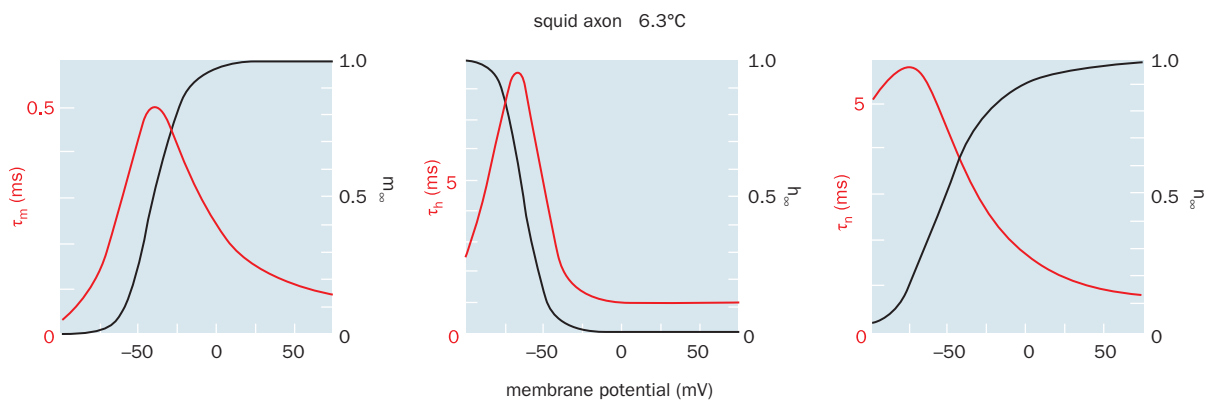


Figure 2.7 The steady-state and time constant as a function of membrane potential V for the gating variables m , h and n of the Hodgkin-Huxley model of an action potential. The steady-state is an increasing function of V for the activation variables m and n of the sodium and potassium conductances, respectively, whereas it is a decreasing function of V for the inactivation variable h of I_{Na} . The time constant for m is about tenfold shorter than those for h and n .

When the input current I_{app} is sufficiently large, above a critical value I_c , the Hodgkin-Huxley neuron model generates periodic discharges of action potentials (Figure 2.8a). The repetitive firing can be intuitively understood as a result of the interplay between fast positive feedback mediated by I_{Na} and slower negative feedback mediated by I_K . Because the activation variable m increases with V (Figure 2.7, left), depolarization leads to the opening of I_{Na} , which as an inward current further increases V . This positive feedback occurs like a chain reaction very quickly (τ_m is sub-millisecond, about 10-fold shorter than τ_h and τ_n , see Figure 2.7), underlying the rapid upstroke of an action potential. On the other hand, the inactivation variable h decreases with V (Figure 2.7, middle). Thus, the sodium conductance decreases on a slower timescale. Moreover, the activation variable n of I_K grows with depolarization (Figure 2.7, right; Figure 2.8a). This outward potassium current eventually produces the downstroke of an action potential. During a refractory period, the membrane potential is hyperpolarized, causing n (thus I_K) to decay away. At the same time, the inactivation variable h for the sodium conductance recovered to a high value. The whole cycle can start anew, leading to a periodic train of action potentials.

Oscillation underlying repetitive firing of action potentials naturally results from a fast positive feedback process and a slower negative feedback process. Indeed, if the negative feedback by activation of g_K was as fast as, or faster than, the positive feedback by activation of g_{Na} , rhythmic action potential would be impossible. Instead, the membrane potential would reach a steady-state regardless of the injected current intensity I_{app} . As a general principle, oscillation occurs in a system endowed with a fast positive feedback (here the interplay between depolarization and activation of I_{Na}) and a slower negative feedback (the back-and-forth between voltage-activated I_K and its hyperpolarizing influence on V).

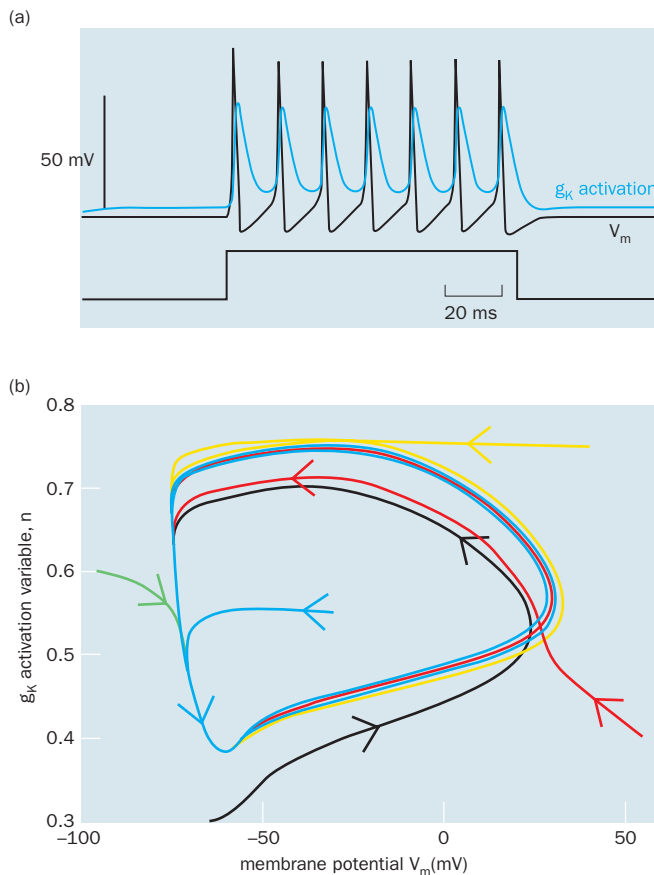


Figure 2.8 Repetitive firing of action potentials in the Hodgkin-Huxley model. (a) Membrane potential V (black) and the activation variable of the potassium conductance n (blue) as a function of time, in response to a long current pulse $I_{app} = 9 \mu A / cm^2$. (b) The potassium activation variable n is plotted against V , showing that the oscillation forms a closed trajectory in this projected two-dimensional “state space.” The trajectory roughly consists of three portions: The upstroke of an action potential (when both V and n increase), the downstroke (when V decreases while n keeps increasing then starts to decrease), and the refractory period (when n continues to decrease while V starts to increase). Different colors correspond to five different initial conditions (with different V and n values at $t = 0$); in all cases the system dynamically evolves into the limit cycle of the oscillatory attractor state.

In the Hodgkin-Huxley model, as the input current I_{app} is increased gradually across I_c , the behavior undergoes a *qualitative* change from steady-state to repetitive firing. Such sudden emergence of a novel behavior can be understood mathematically in the theory of nonlinear dynamical systems as a *bifurcation* phenomenon, and I_c is called a bifurcation point of the parameter I_{app} (Strogatz 2016). Importantly, bifurcations can never occur in linear dynamical systems such as an RC circuit.

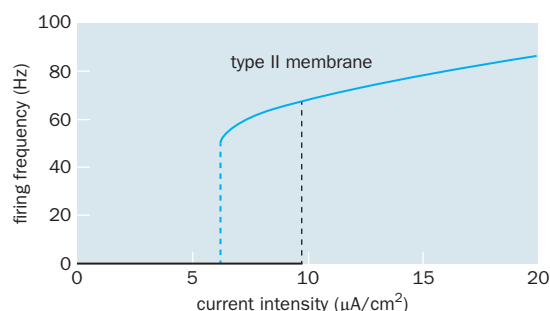
Mathematically, a stable state is called an *attractor*, which simply means that when the system deviates slightly away from that state, for instance, by a small perturbation, it will evolve back to that state over time. We have seen the steady-state of an RC circuit as a stationary attractor. On the other hand, a regular train of action potentials in the Hodgkin-Huxley model represents an oscillatory type of attractor behavior called *limit cycle*. The attractor nature of periodic spiking is shown in **Figure 2.8b**: regardless of the initial states, the model always converges to the same periodic attractor state. If the system is perturbed by a transient stimulus, it would evolve over time and eventually resume the same firing pattern after the stimulus offset, except for a shift of spiking time, or the phase of the periodic attractor state. Thus, a periodic attractor is robust (the amplitude and periodicity are not sensitive to transient perturbations). At the same time, the phase (e.g., the time relative to the peak of an action potential in every oscillatory cycle) can be readily shifted by an external stimulation, hence the clock can be readily reset.

2.3.2 Type I and Type II Neurons

Figure 2.9 displays the frequency-current function for the classic Hodgkin-Huxley model. Contrary to the LIF model, in the Hodgkin-Huxley model the firing rate of action potentials is not zero but finite (~ 45 Hz) at the threshold input current, and the range of possible firing rates is rather limited. A finite minimum firing rate at the onset of repetitive spike firing is a characteristic of the current-frequency relationship for “type II” neurons (Hodgkin 1948; Rinzel and Ermentrout 1989; Wang and Rinzel 1995).

This minimum firing rate of ~ 45 Hz reflects an intrinsic oscillatory frequency even when V is subthreshold. Indeed, when the input current I_{app} is near the firing threshold, the membrane potential displays damped temporal oscillations around its steady-state ($V_{ss} \simeq -60$ mV), just like a pendulum in a viscous fluid swinging back to its stable equilibrium state. The damped oscillation is, again, produced by the interplay between the fast positive feedback and slow negative feedback mediated by I_{Na} and I_K , as explained previously for the action potential generation; but when the input current is not sufficiently strong to drive V above the spiking threshold, oscillations are subthreshold and occur on a small scale of membrane potential variations around -60 mV. Assuming the positive feedback is very fast, the upswing and downswing of oscillation are roughly given by the time it takes for the negative feedback to build up and then decay away. The time

Figure 2.9 Frequency of repetitive firing as a function of input current intensity of the Hodgkin-Huxley model. Note a gap of firing rate between 0 and ~ 45 Hz and a range of I_{app} where bistability exists between a stable steady-state (with zero firing rate) and an oscillatory attractor of repetitive firing for I_{app} between 7 and 10.



constants of the activation of I_K and inactivation of I_{Na} are $\sim 5 - 10$ ms (**Figure 2.7**), yielding a periodicity of ~ 20 ms in agreement with a frequency of 45 Hz. This intrinsic frequency sets a minimum for firing rate of action potentials when the input current exceeds the spiking threshold, thereby creating a gap in the frequency-current curve (**Figure 2.9**).

In contrast to the classic Hodgkin-Huxley model, many neurons in the mammalian cortex exhibit a continuous current-frequency curve, starting at zero firing rate ("Type I neuron"), and the firing rate can be as high as a few hundred hertz (McCormick et al. 1985; Koch 1999). It turns out that a few quantitative changes of the parameters from the original Hodgkin-Huxley model gives rise to "Type I" behavior (Wang and Buzsáki 1996). These changes involve shifting the activation and inactivation curves of I_{Na} and I_K to more depolarized membrane potential, by a few mV for m_∞ and h_∞ and by more than 20 mV for n_∞ . As a result, the voltage-dependent conductances are no longer significant at ~ -60 mV. Furthermore, because the activation of I_K requires a lot more depolarization compared to I_{Na} , once activated the positive feedback mediated by I_{Na} inevitably produces sufficient depolarization for the generation of an action potential. Therefore, subthreshold oscillation is no longer possible, and spike firing starts at zero hertz (**Figure 2.10**). Finally, the gating kinetics of I_{Na} and I_K speed up, producing a broad range of firing rates (from 0 to 400 Hz).

The ion channel model of type I neurons (Wang and Buzsáki 1996) can be captured by a modified integrate-and-fire model. Fourcaud-Trocme et al. (2003) introduced an exponential term into the LIF equation that describes the positive feedback underlying action potential. The exponential integrate-and-fire (EIF) model obeys

$$c_m \frac{dV}{dt} = -g_L(V - V_L) + \Delta \exp((V - V_{th}) / \Delta) + I_{app} \quad (2.21)$$

where V_{th} is the voltage firing threshold, and Δ measures the sharpness of the action potential upstroke. Thus, when V approaches V_{th} , the inward current grows exponentially with V , providing a rapid and powerful positive feedback. Each time V reaches V_{th} , a spike is triggered and V is reset to V_{reset} for a time that includes the duration of an action potential (τ_{spike}) and a refractory period (τ_{ref}). With the appropriate choice of model parameter

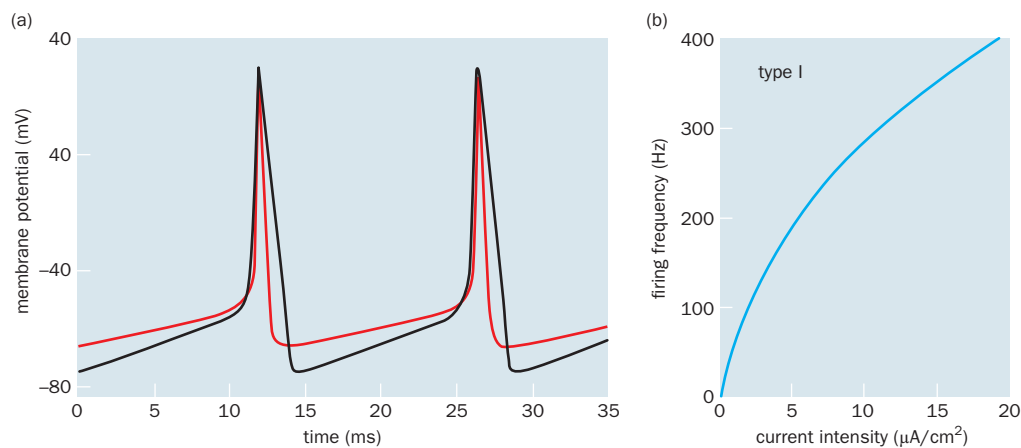


Figure 2.10 Frequency of repetitive firing as a function of input current intensity of a type I neuron. (a) Compared to the original Hodgkin-Huxley model (black), a modified Hodgkin-Huxley model displays narrower spikes and less afterhyperpolarization (red). (b) The firing rate ranges from 0 to about 400 Hz. Reproduced from Wang and Buzsáki (1996). [Wang, X. J., & Buzsáki, G. (1996). Gamma oscillation by synaptic inhibition in a hippocampal interneuronal network model. *Journal of Neuroscience*, 16(20), 6402–6413.]

values, the EIF model reproduces quite accurately the behavior of the Wang-Buzsaki model. In particular, the current-frequency relationship is virtually indistinguishable between the two models. In the absence of noise, for input current near the firing threshold value I_c , the firing rate is predicted to behave as $r = \beta \sqrt{I_{app} - I_c}$ with $\beta = 38 \text{ Hz}/((\mu\text{A})^{1/2}/\text{cm})$ (Fourcaud-Trocme et al. 2003). As in the classical LIF, with the addition of a white noise to the input, neuronal firing can be triggered by fluctuating inputs for I_{app} below I_c , and the current-frequency curve of the type I neuron model is again reproduced by the EIF model.

2.4 TIME-DEPENDENT NEURONAL FIRING PATTERNS

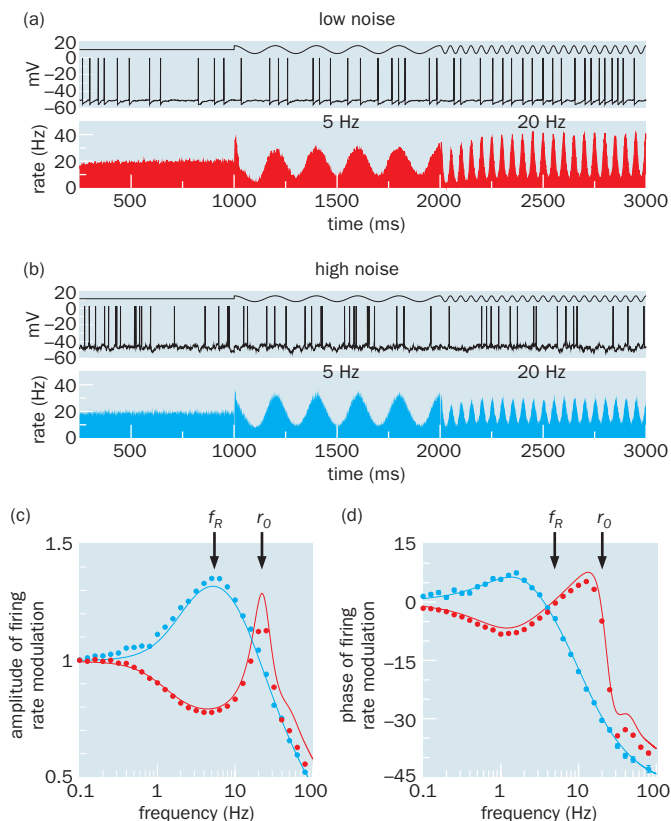
2.4.1 Resonance in Response to Time-Dependent Noisy Inputs

The frequency-current relationship describes the neuronal response to a constant input (or a stationary input in the presence of noise). How does a neuron respond to a time-varying input? For instance, can a neuron “keep up” with increasingly rapid input fluctuations? When synaptic inputs reflect a synchronous network rhythm at frequency f , can a neuron respond effectively at a higher and higher frequency?

To address these questions, consider a noisy oscillatory input current $I_{app} = I_0 + I_1 \cos(2\pi ft) + I_{noise}$, where I_0 sets the overall firing rate above 0 (say 20 Hz), I_1 is the amplitude of the sinusoidal input and I_{noise} is a white noise. **Figure 2.11** shows an integrate-and-fire neuron model generalized to include a slow outward current (Richardson et al. 2003). In response to a noisy sinusoidal input with a low (a) or high (b) noise level (upper panel in [a] and [b]), membrane potential fluctuates over time (middle panel in [a] and [b]), and spike train varies considerably from trial to trial. Across trials

Figure 2.11 Subthreshold and spiking resonance of a model neuron in the presence of noise. The model neuron exhibits subthreshold membrane resonance at frequency $f_R \approx 5 \text{ Hz}$.

(a) Response with low noise. Top: Input current with three (tonic, 5 Hz and 20 Hz sinusoidal current injection) epochs. The mean output firing rate remains constant ($r_0 = 20 \text{ Hz}$) throughout. Middle: A sample membrane trace. Bottom: Post-stimulus time histogram (PSTH). (b) Response with high noise. Same format as in (a). (c) Signal gain amplitude versus input frequency f (normalized to be one at $f = 0$). Firing rate resonance occurs at the subthreshold resonance frequency ($f \approx f_R$) with high noise (blue), but at the mean firing frequency ($f \approx r_0$) with low noise (red). (d) Phase shift of the firing response relative to the sinusoidal input (positive: Phase advance; negative: Phase delay). Panels (a)–(c) are reproduced from Richardson et al. (2003). [Richardson, M. J., Brunel, N., & Hakim, V. (2003). From subthreshold to firing-rate resonance. *Journal of Neurophysiology*, 89(5), 2538–2554.



one can compute the number of spikes within each small time window, normalized by the time window size, to yield the trial-averaged firing rate $r(t)$ (post-stimulus time histogram, or PSTH) (bottom panel in [a] and [b]).

Under the condition that that rhythmic modulation is small relative to the baseline (I_1 is small compared to I_0), $r(t)$ can be described as $r(t) = r_0 + r_1(f) \cos(2\pi ft + \Phi(f))$, where $r_1(f)$ is the amplitude of rhythmic modulation and $\Phi(f)$ is a phase shift relative to the input current. At low f , the neuronal response can “keep up” with input oscillations and $r_1(f)$ is roughly independent of f . By contrast, for f above a “cutoff” value f_c , $r_1(f)$ decreases with f , reflecting the fact that neuron dynamics cannot follow effectively more rapid changes in the input. Both the type I neuron (Wang and Buzsáki 1996) and its reduced EIF model (Richardson et al. 2003) predict that $r_1(f)$ decays with f as $1/f$, whereas the phase shift $\Phi(f)$ reaches an asymptotic level of 90 degrees at high frequencies (Fourcaud-Trocme et al. 2003). In other words, neural firing activity lags input current by up to a quarter of a cycle. By contrast, for the classical LIF model driven by a sinusoidal input and white noise, the attenuation of firing response decreases less steeply, as $r_1(f) \sim 1/\sqrt{f}$, and $\Phi(f)$ plateaus at 45 degrees. Therefore, even though LIF displays a current-frequency relationship with zero onset rate, its behavior in response to temporally varying noisy inputs differs qualitatively from the EIF model and the ion channel model of type I neurons.

Some types of neurons display a resonance phenomenon, where the oscillatory component of neural responses $r_1(f)$ shows a peak at a “preferred” frequency. As shown in **Figure 2.11c**, at low noise (red) the resonance frequency is the mean firing rate r_0 . On the other hand, at high noise (blue) the resonance frequency is determined by damped oscillations of subthreshold membrane potential, which is around 5 Hz in this case. Interestingly, around the resonance frequency, neural activity shows a phase advance rather than a delay with respect to the input current; i.e., it tends to peak on the upswing of the oscillatory input (**Figure 2.11c-d**) (Kepecs et al. 2002; Richardson et al. 2003).

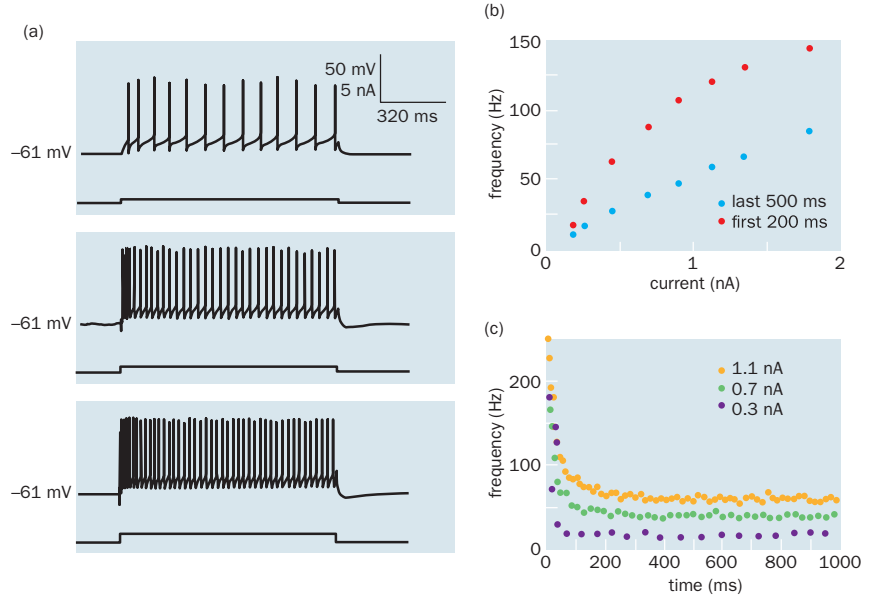
2.4.2 Spike Rate Adaptation

In addition to the transient sodium current and non-inactivating potassium current that underlie action potential generation, neurons display a variety of other active ion currents that are gated by membrane potential or intracellular calcium ions (Llinás 1988; Koch 1999). Those currents are expressed differentially in different types of neurons, endowing them with a rich repertoire of membrane potential response dynamics and spike firing patterns. A particularly prominent example is spike rate adaptation (SRA), commonly observed in a majority of excitatory pyramidal cells and some inhibitory neurons of the mammalian cortex (McCormick et al. 1985; Lorenzon and Foehring 1992; Markram et al. 2004). SRA refers to the phenomenon of a gradual decrease of spiking discharges of a neuron in response to a long-lasting stimulus, as a form of neuronal dynamics that renders a single neuron to be more sensitive to changes in the input rather than steady-state, in order to achieve efficient coding of inputs. SRA can be quantified by the inverse of instantaneous ISI as a function of time, $r(t) = 1/ISI(t)$ (**Figure 2.12**). It was found both *in vitro* (Lorenzon and Foehring 1992) and *in vivo* (Ahmed et al. 1998) that $r(t)$ can be well fit with an exponential function of time,

$$r(t) = r_{ss} + (r_0 - r_{ss}) \exp(-t / \tau_{adap}) \quad (2.22)$$

The degree of adaptation is quantified by $F_{adap} = (r_0 - r_{ss}) / r_0$, which is 0 without adaptation ($r_{ss} = r_0$), and 1 with maximum adaptation (in the case

Figure 2.12 Spike frequency adaptation of an excitatory pyramidal neuron from the human cortex recorded in *in vitro* slices. (a) Membrane potential in response to a current injection at three intensity values. (b) The initial firing rate and steady-state firing rate as a function of input current intensity. (c) The instantaneous firing rate (defined as the inverse of an interspike interval) as a function of time for the three examples from (a), each can be fitted with an exponential function of time. Reproduced from Lorenzon and Foehring (1992). Lorenzon, N. M., & Foehring, R. C. (1992). Relationship between repetitive firing and afterhyperpolarizations in human neocortical neurons. *Journal of Neurophysiology*, 67(2), 350–363.



of $I_{SS} = 0$). Therefore, SRA can be fully described with two quantities, F_{adap} and τ_{adap} . Physiological studies have revealed that, in cortical pyramidal neurons, SRA on a timescale of a few hundreds of milliseconds is primarily mediated by a calcium-activated potassium ion current (I_{KCa}) (Sah 1996), one of many types of potassium currents (Rudy 1988). I_{KCa} is sometimes also denoted as I_{K-AHP} because it underlies slow afterhyperpolarization (AHP). The way I_{KCa} works is that membrane depolarization during spikes induces calcium influx in the cell through an “L-type” calcium channel I_{Ca-L} . The intracellular calcium level $[Ca^{2+}]$ (relative to a low baseline) accumulates over time and activates I_{KCa} . This outward current reduces the neuronal excitability as a negative feedback mechanism. Thus, the firing rate decreases over time.

SRA can be incorporated in the LIF model by assuming $I_{KCa} = G_{KCa}[Ca^{2+}](V - V_K)$, where $V_K = -80$ mV is the reversal potential. A modified integrate-and-fire model with adaptation is given by (Liu and Wang 2001):

$$C_m \frac{dV}{dt} = -G_L(V - V_L) - G_{KCa}[Ca^{2+}](V - V_K) + I_{app} \quad (2.23)$$

$$\frac{d[Ca^{2+}]}{dt} = \alpha \sum_j \delta(t - t_j) - \frac{[Ca^{2+}]}{\tau_{Ca}} \quad (2.24)$$

where $\{t_j\}$ is a presynaptic spike train, I_{Ca-L} mediated calcium influx per spike is described by a delta-function, α is the amount of calcium influx per spike (~ 200 nM), and τ_{Ca} is the decay time constant (~ 100 ms) (Helmchen et al. 1996).

An example of spike train generated by this adaptive LIF model is shown in **Figure 2.13**, where the stimulation is a stochastic Poisson input. The resulting firing rate time course $r(t)$ is well fit by Eq. (2.22) with $F_{adap} = 0.7$ and $\tau_{adap} = 30$ ms, similar to *in vivo* observation of cortical neurons (Ahmed et al. 1998).

How are F_{adap} and τ_{adap} determined by the biophysical parameters, and why is τ_{adap} much shorter than τ_{Ca} even though adaptation results from the calcium dynamics? To answer these questions, a theoretical analysis can be carried out in several steps (Wang 1998; Liu and Wang 2001).

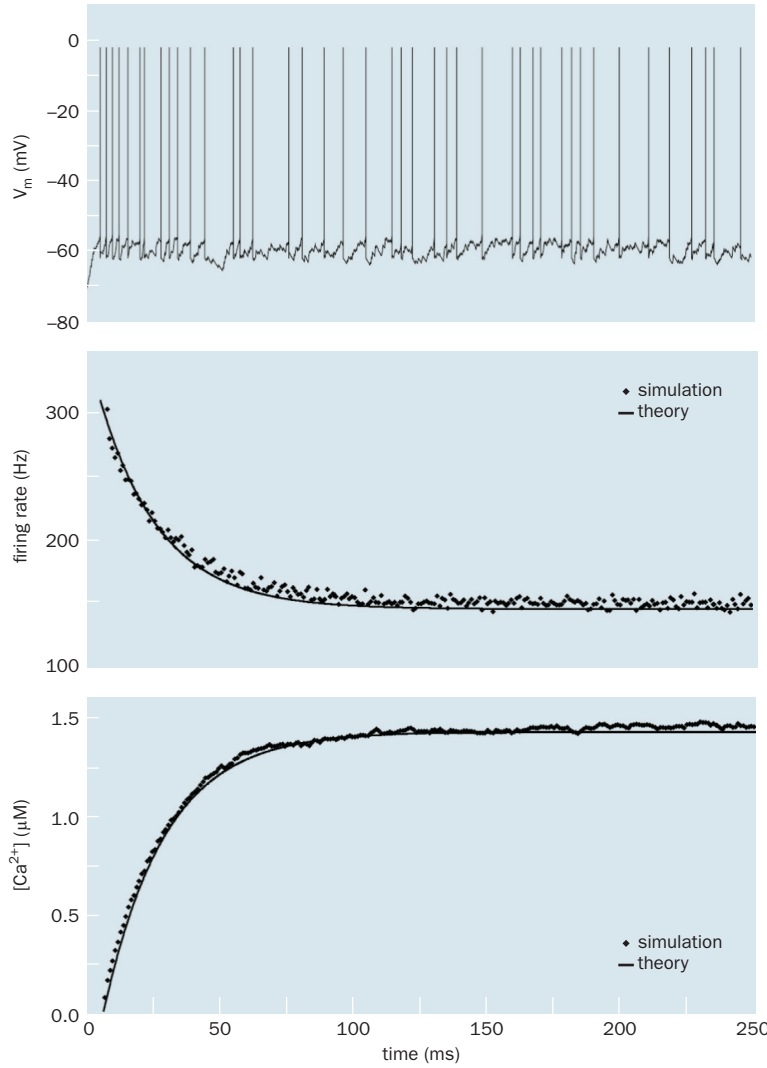


Figure 2.13 Adaptation dynamics of an adaptive LIF model neuron. Upper panel: Adaptation in response to Poisson synaptic inputs. Middle panel: The trial-averaged instantaneous firing rate decreases in time exponentially with a characteristic time constant τ_{adap} . Lower panel: The intracellular $[Ca^{2+}]$ level builds up with the same time course. Adapted with permission from Liu and Wang (2001). [Liu, Y. H., & Wang, X. J. (2001). Spike-frequency adaptation of a generalized leaky integrate-and-fire model neuron. *Journal of Computational Neuroscience*, 10(1), 25–45.]

First, let us assume that the firing rate $r(t) = F(I_{tot}) \simeq \beta I_{tot}$ (approximately linear with the total current $I_{tot} = I_{app} - I_{KCa}$, above the firing threshold). Second, replacing V by a constant average $\langle V \rangle$ in I_{KCa} :

$$r(t) \simeq \beta (I_{app} - G_{KCa} [Ca^{2+}] (\langle V \rangle - V_K)) \simeq r_0 - J_{KCa} [Ca^{2+}] \quad (2.25)$$

where $r_0 = \beta I_{app}$ is the initial firing rate (without adaptation) and $J_{KCa} = \beta G_{KCa} (\langle V \rangle - V_K)$ quantifies the effective strength of the adaptation current. Third, assuming that electrical activity is faster than calcium signal, which thus averages spikes over time, so that the spike train in the $[Ca^{2+}]$ equation can be approximately replaced by the firing rate r ,

$$\frac{d[Ca^{2+}]}{dt} = \alpha \sum_j \delta(t - t_j) - [Ca^{2+}] / \tau_{Ca} \quad (2.26)$$

$$\simeq \alpha r - [Ca^{2+}] / \tau_{Ca} = \alpha r_0 - [Ca^{2+}] / \tau_{adap} \quad (2.27)$$

with

$$\frac{1}{\tau_{adap}} = \alpha J_{KCa} + \frac{1}{\tau_{Ca}} \quad (2.28)$$

The linear Eq. (2.27) can be readily solved, yielding

$$[Ca^{2+}](t) = [Ca^{2+}]_{ss} (1 - \exp(-t / \tau_{adap})) \quad (2.29)$$

with $[Ca^{2+}]_{ss} = \alpha r_0 \tau_{adap}$. This combined with Eq. (2.25) yields

$$r(t) = r_{ss} + (r_0 - r_{ss}) \exp(-t / \tau_{adap}) \quad (2.30)$$

with $r_{ss} = r_0 - J_{KCa} [Ca^{2+}]_{ss}$.

This theory fits the numerical results quite well. In particular, it explains why τ_{adap} is typically much shorter than τ_{Ca} . For instance, if $\alpha J_{KCa} = 0.02$ and $\tau_{Ca} = 100$ ms ($1 / \tau_{Ca} = 0.01$) and we have $\tau_{adap} \simeq 30$ ms. More surprisingly, it predicts a linear relationship between F_{adap} and τ_{adap} . According to Eq. (2.27), $[Ca^{2+}]_{ss}$ can be written in two different ways,

$$\alpha r_{ss} - [Ca^{2+}]_{ss} / \tau_{Ca} = \alpha r_0 - [Ca^{2+}]_{ss} / \tau_{adap} = 0 \quad (2.31)$$

Since $[Ca^{2+}]_{ss}$ from the two equations must be the same, we deduce

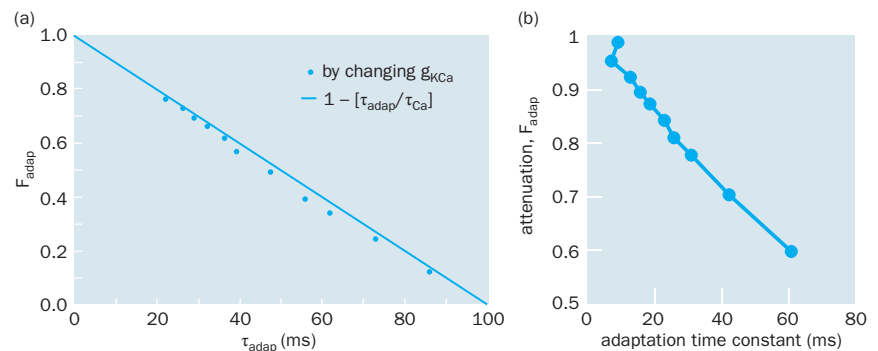
$$F_{adap} = 1 - \frac{\tau_{adap}}{\tau_{Ca}}. \quad (2.32)$$

Thus, there is a simple relationship between F_{adap} and τ_{adap} that only depends on a biophysical parameter (the effective time constant of calcium extrusion τ_{Ca}) (Wang 1998). Eq. (2.32) predicts that stronger adaptation has faster dynamics: F_{adap} increases linearly with decreasing τ_{adap} , as the intensity of input current I_{app} or the adaptation conductance G_{KCa} is varied (Figure 2.14a). This theoretical prediction was confirmed by a physiological experiment (Gabbiani and Krapp 2006) (Figure 2.14b).

2.4.3 Input Decorrelation

SRA takes place on multiple timescales (Marom 2010; Weber et al. 2019). A slower form of adaptation in a single neuron is realized by a potassium current activated by the intracellular sodium concentration rather than by calcium concentration (Sanchez-Vives and McCormick 2000; Sanchez-Vives et al. 2000). What happens is that prolonged firing of action potentials leads to a gradual Na^+ influx into the cell through the opening of sodium channels. Intracellular sodium accumulation eventually activates a potassium current, I_{KNa} , which gives rise to hyperpolarization of the neuronal membrane and a reduced responsiveness of the cell. After the offset of a large stimulus, sodium ions are slowly extruded from the cell through a Na^+/K^+ ionic pump. This process underlies the recovery of neuronal responsiveness with a time constant of about ten seconds.

Figure 2.14 A relationship between the amount of adaptation F_{adap} and adaptation time constant τ_{adap} of spike rate adaptation by a single neuron. Left: Model prediction; right: Experimental measurement in a looming sensitive neuron. Panel (a) is reproduced from Liu and Wang (2001), (b) from Gabbiani and Krapp (2006). [Liu, Y. H., & Wang, X. J. (2001). Spike-frequency adaptation of a generalized leaky integrate-and-fire model neuron. *Journal of Computational Neuroscience*, 10(1), 25–45.] [Gabbiani, F., & Krapp, H. G. (2006). Spike-frequency adaptation and intrinsic properties of an identified, looming-sensitive neuron. *Journal of Neurophysiology*, 96(6), 2951–2962.]



This phenomenon, observed at the level of single cells, is directly relevant to perception of an animal. Our visual system has evolved to be more sensitive to changes in the visual world than stimuli that are constant in time. For instance, when we enter a dark room from a bright one, we momentarily become blind, but as our eyes adjust to the dim light, we start to see things. We also adapt to contrast, which is the difference in luminance of an object relative to its surroundings; thus, a face of the same luminance appears brighter indoor than under the sun in a park.

In neurophysiological laboratory experiments, contrast of a visual stimulus, such as sinusoidal grating, could be varied, and it was found that spike firing rate of V1 single cells display adaptation on a timescale of seconds comparable to contrast adaptation at the behavioral level (Ohzawa et al. 1985; Sanchez-Vives and McCormick 2000). *In vitro* experiments on single cells suggested that a sodium activated potassium current I_{KNa} could account for contrast adaptation (Sanchez-Vives et al. 2000).

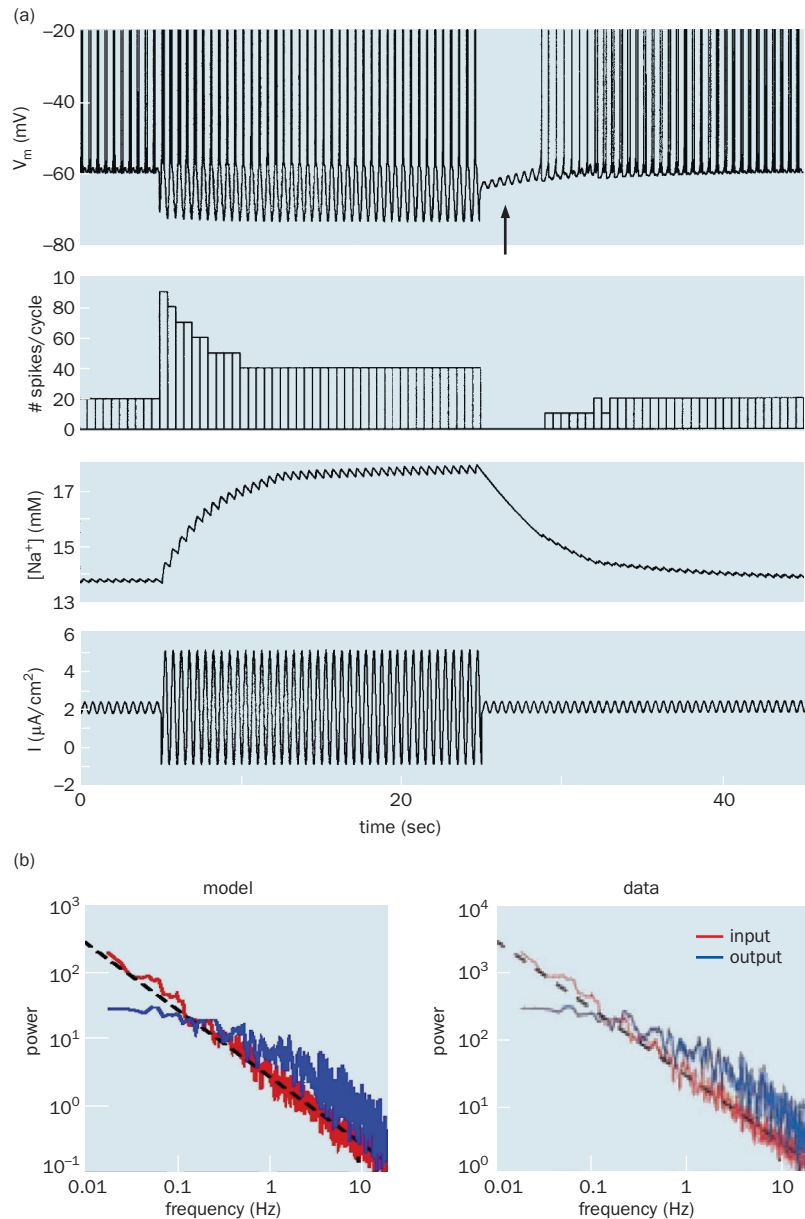
Figure 2.15 displays a type I neuron model with the addition of an I_{KNa} , and intracellular Na^+ dynamics (influx through I_{Na} and outflux through a Na^+/K^+ pump) (Wang et al. 2003). The input is as in the experimental protocol, namely, a sinusoidal current (at 2 Hz) mimicking a sinusoidal grating (Ohzawa et al. 1985; Sanchez-Vives and McCormick 2000). Input amplitude represents stimulus contrast. Initially, the model neuron responds to a low amplitude input at a constant rate. During a high amplitude input, the firing rate transiently increases, then decays to a lower steady-state level according to a time course in parallel with that of intracellular Na^+ accumulation, hence activation of I_{KNa} . After the stimulus returns from the high to the original low amplitude, the membrane potential is hyperpolarized, and the model neuron is incapable of spiking response for seconds. Slowly, however, intracellular Na^+ decays back to baseline, and the model neuron resumes the same activity level as before. Thus, the model captures essential observations from neurons in the primary visual cortex.

What may be the behavioral impact of such neuronal adaptation? Sensory inputs are highly redundant. Removal of redundancy has long been proposed to enhance the efficiency of neural coding of sensory stimuli in the brain, and neuronal adaptation provides a mechanism for a decorrelation operation that reduces input redundancy in time (Barlow 1961; Barlow and Foldiak 1989).

If I_{KNa} subserves adaptation, is it capable of input decorrelation as well? Specifically, light intensity signals of optical flow, observed with a narrow visual field, exhibit correlations at all timescales. This is quantified by a scale-free power spectrum, with the power growing like $1/f$ at decreasing frequencies f (van Hateren and van der Schaaf 1996) (red curve in **Figure 2.15b**), characterizing correlations at increasingly longer timescales. By contrast, a completely uncorrelated signal has a constant power spectrum, i.e., the same power at all frequencies (hence the name “white noise”). How would a cell endowed with adaptation respond to an $1/f$ -type input? This question can be investigated in model simulations, using stochastic inputs with strong time correlations. When an $1/f$ stochastic input is used as injected current to drive the model neuron, long-term correlations of the spiking activity are suppressed, so that the power spectrum of the output becomes essentially flat (“whitened”) at frequencies below 1 Hz (blue curve, left panel in **Figure 2.15b**). Decorrelation takes place due to the negative feedback mediated by I_{KNa} : a higher firing activity leads to a larger I_{KNa} , hence a stronger negative signal to subtractively counterbalance the external input drive. Decorrelation takes place on a timescale of a second or longer, hence whitening of the power spectrum below 1 Hz. Therefore, a single neuron mechanism can subserve decorrelation function. This model prediction was directly tested in

Figure 2.15 Adaptation and decorrelation by a single neuron.

(a) The response of the model neuron to a low-high-low sinusoidal current input. Top to bottom: Membrane potential, number of spikes per cycle, intracellular Na^+ concentration, and 2 Hz sinusoidal injected current that varies from low to high to low in amplitude. Following the injection of high-amplitude sinusoidal current, there is a slow hyperpolarization lasting for about ten seconds (indicated by an arrow). Intracellular Na^+ accumulation mirrors the adaptation time course of the instantaneous firing rate during the high-contrast input, whereas Na^+ decay is correlated with the slow recovery from hyperpolarization during the second low-amplitude period. (b) Decorrelation of $1/f$ -type stochastic input by the model (left panel) and a visual cortical neuron recorded intracellularly in a ferret cortical slice (right panel). Reproduced from Wang et al. (2003). [Wang, X. J., Liu, Y., Sanchez-Vives, M. V., & McCormick, D. A. (2003). Adaptation and temporal decorrelation by single neurons in the primary visual cortex. *Journal of Neurophysiology*, 89(6), 3279–3293.]



an *in vitro* preparation of ferret cortical slices. When a real neuron of the primary visual cortex receives a stochastic $1/f$ -type input, the output of the neuron exhibits the same kind of decorrelation as the model (Figure 2.15b, right panel). This experiment demonstrates that decorrelation can occur in single cortical cells.

2.5 BURST FIRING

2.5.1 Ping-Pong Interplay between Soma and Dendrite

Compartmental modeling of dendrites (Segev et al. 1995) reduces a spatially continuous dendrite into a number of discrete compartments, each is assumed to be isopotential. The interactions between two compartments j and k are mediated by a current flow through an axial resistance R_{jk} , and act to equalize the voltages V_j and V_k . The smaller the resistance R_{jk} , the more effective the current flow and the inter-compartment coupling. In principle, the value of R_{jk} (the coupling strength) may be different for any

pair (j and k) of compartments, because it depends on the physical properties such as the radius and length of each cylinder-shaped compartment (Koch 1999; Dayan and Abbott 2001).

Importantly, dendrites are not passive but endowed with various types of active ion channels and show a rich repertoire of electroresponsiveness (Stuart et al. 2016). In particular, voltage-dependent ion channels at dendritic sites, once activated, last for tens of milliseconds, and this plateau potential provides a depolarizing drive during a burst firing of multiple spikes (Wong et al. 1979; Traub et al. 1991; Pinsky and Rinzel 1994; Wang 1999a; Kepecs et al. 2002). Furthermore, Hodgkin-Huxley-type sodium and potassium currents on the dendrite enable action potential (initiated at the initial segment of axon, near the soma) to propagate back into the dendrite (Stuart et al. 1997). Therefore, there is a soma-dendritic “ping-pong” interplay: an action potential backpropagates and depolarizes dendritic membrane potential, activates dendritic voltage-gated channels and prolongs dendritic depolarization, which in turn produces more spikes. In the neocortex, such burst firing is especially prominent in pyramidal cells of deep layers (Silva et al. 1991).

Since burst firing is rather common in neural systems, let us discuss it in some detail. We shall use a relatively simple model that captures the ping-pong mechanism (Pinsky and Rinzel 1994), which is deduced from a more elaborated model with hundreds of compartments (Traub et al. 1991). Consider a two-compartment model with a soma and a dendrite (**Figure 2.16a**). The somatic voltage V_s and dendritic voltage V_d obey the following current-balance equations:

$$c_m A_s \frac{dV_s}{dt} = -A_s (I_L + I_{Na} + I_K + I_s) - \frac{1}{R_c} (V_s - V_d); \quad (2.33)$$

$$c_m A_d \frac{dV_d}{dt} = -A_d (I_L + I_{Ca-L} + I_{K-C} + I_{KCa} + I_d) - \frac{1}{R_c} (V_d - V_s) \quad (2.34)$$

where $c_m = 1 \mu\text{F}/\text{cm}^2$. The membrane surface areas are A_s and A_d (in μm^2), and the axial resistance (resistive coupling) is denoted as R_c (in $\text{M}\Omega$). The soma contains Hodgkin-Huxley-type spike-generating currents I_{Na} and I_K (see Section 2.3). The dendrite contains three active conductances: a voltage-gated, L-type calcium current I_{Ca-L} , a calcium-activated potassium current I_{KCa} (the same current I_{KCa} underlying spike-rate adaptation [Section 2.4]), and another potassium current I_{K-C} , which depends on both dendritic voltage and intracellular calcium concentration. The injected currents to the perisomatic and dendritic compartments are I_s and I_d (all currents are in $\mu\text{A}/\text{cm}^2$),

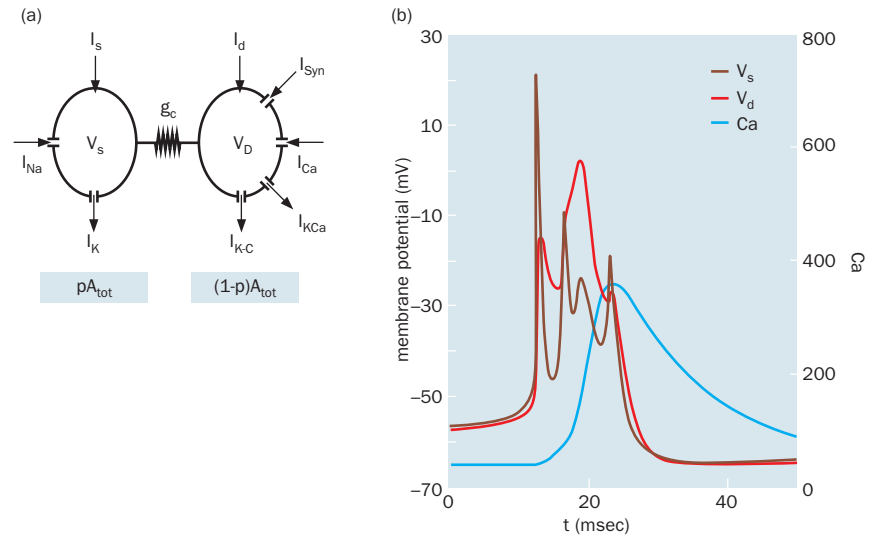
Defining the total membrane area $A_{\text{tot}} = A_s + A_d$, and the fractional area for the perisomatic membrane $p = A_s / A_{\text{tot}}$, one can rewrite the two equations as

$$c_m \frac{dV_s}{dt} = -(I_L + I_{Na} + I_K + I_s) - \frac{g_c}{p} (V_s - V_d); \quad (2.35)$$

$$c_m \frac{dV_d}{dt} = -(I_L + I_{Ca-L} + I_{K-C} + I_{KCa} + I_d) - \frac{g_c}{(1-p)} (V_d - V_s) \quad (2.36)$$

where $g_c = 1 / (R_c A_{\text{tot}})$ (in mS/cm^2). This form of equations helps us to see better the interactions between somatic and dendritic membranes. The resistive coupling acts to equalize V_s and V_d : if V_d is higher than V_s , then the coupling term in the first equation is positive, whereas it is negative in the second equation, thereby acting to increase V_s and decrease V_d . The coupling term is zero when $V_s = V_d$. Moreover, soma-dendritic interactions are not symmetrical: if the dendritic compartment is larger than the somatic compartment, p is smaller than 0.5, g_c / p is larger $g_c / (1-p)$,

Figure 2.16 Bursting in a two-compartmental model of pyramidal neuron. (a) Schematic of the two-compartment model showing applied currents and outward and inward active currents to soma and dendrite compartments. Soma has Hodgkin-Huxley type sodium (I_{Na}) and potassium (I_K) currents; dendrite has a high-threshold L-type calcium current (I_{Ca-L}), Ca^{2+} - and voltage-activated potassium (I_{K-C}), potassium after hyperpolarization (I_{KCa}) currents. I_s and I_d are applied currents, with the addition of a synaptic current (I_{syn}). Coupling strength between soma and dendrite is governed by parameter g_c and A_{tot} refers to the total cell membrane area. (b) Time courses of somatic voltage, V_s , dendritic voltage, V_d , and Ca during a burst (the scale for Ca is different from voltage). Burst is initiated by a somatic action potential, which triggers a subthreshold dendritic Ca spike by I_{Ca-L} . This is followed by a full Ca spike in the dendrite, leading to somatic burst pattern. Dendritic spike, and hence burst, is terminated by I_{K-C} and I_{KCa} , which build up when Ca reaches appreciable levels. Reproduced from Pinsky and Rinzel (1994). [Pinsky, P. F., & Rinzel, J. (1994). Intrinsic and network rhythmogenesis in a reduced Traub model for CA3 neurons. *Journal of Computational Neuroscience*, 1(1), 39–60.]



reflecting the intuition that there is a stronger influence of the larger dendritic compartment to the smaller somatic compartment than in the opposite direction.

Figure 2.16b displays a burst of spikes produced by this model. During a burst, action potentials originate in the perisomatic compartment (V_s , brown line) and backpropagate into the dendritic compartment (V_d , dashed line) (Larkum et al. 1999). This interaction is described as a dynamic “ping-pong” flow of current between the soma and the dendrite. A burst occurs when a narrow somatic spike (in V_s) elicits a broad dendritic spike (in V_d , red line) that, in turn, produces a depolarizing afterpotential in the soma. The broad dendritic spike, which is due to a dendritic calcium current I_{Ca-L} , acts as a source of inward current to depolarize the soma and bring it to threshold again. During repetitive spikes, intracellular calcium accumulates ($[Ca^{2+}]$, blue), the resulting gradual activation of the potassium currents I_{K-C} and I_{KCa} eventually shut off local regeneration of dendritic spikes and a burst is terminated. During hyperpolarization $[Ca^{2+}]$ decays back to a baseline, the membrane potential recovers, possibly leading to another burst of spikes.

The two-compartment model describes how a ping-pong mechanism produces bursting firing. However, note that multiple mechanisms contribute to bursting, especially the voltage-gated synaptic conductance mediated by the N-methyl-D-aspartate (NMDA) receptors (which we will discuss in Section 2.6) (Larkum et al. 1999; Larkum 2013).

A two-compartment model may not be suitable for investigating other questions. For instance, relatively independent segments of a dendritic tree suggest a two-layered model of pyramidal cells (Poirazi et al. 2003). To describe input and output control of pyramidal cells by SST/CB-expressing and PV-expressing inhibitory neurons, respectively, a better yet minimal model for pyramidal cells may have three compartments (soma, proximate and distal dendrites), so that the distal dendrite is separated by the proximate dendrite from the soma and can act less dependently (Wang et al. 2004). As another example, the alignment of pyramidal dendrites is an important factor for summing the extracellular current flow across a population of pyramidal cells, which underlies local field potential (LFP), electroencephalography (EEG) and magnetoencephalography (MEG) (Wang 2010; Buzsaki et al. 2012). LFP is captured in a model where pyramidal cells have a more complicated dendrite (Sherman et al. 2016; Jones et al. 2009). Modeling of dendrites is an active research topic

that continues to yield surprising findings (Stuart et al. 2016; Poirazi and Papoutsi 2020; Otor et al. 2022).

2.5.2 Postinhibitory Rebound

Like potassium currents, there are multiple types of calcium currents (Tsien et al. 1987). Another kind of burst firing relies on a low-threshold (T-type) calcium current I_{Ca-T} rather than a high-threshold calcium current I_{Ca-L} , and thus can be activated at membrane potentials below the spiking threshold (Llinás 1988; Wang et al. 1991; McCormick and Huguenard 1992; Wang 1994; Destexhe et al. 1996). Curiously, I_{Ca-T} activation is conditioned on a prolonged membrane hyperpolarization: its inactivation variable h has a steady-state that is virtually zero near or above the resting membrane potential; it increases when V is hyperpolarized with a time constant of ~ 100 ms. On the other hand, its activation variable becomes significant with moderate depolarization. Therefore, when a neuron is endowed with a large I_{Ca-T} , depolarization from resting-state leads to repetitive spikes (tonic mode). On the other hand, upon release from a sufficiently long hyperpolarization, I_{Ca-T} is activated to underlie a “depolarization wave” (postinhibitory rebound) with a burst of spikes riding on top of it (burst mode) (Figure 2.17a). This phenomenon has been observed in several neural types, including thalamic neurons that relay information from sensory organs to the neocortex (Llinás 1988). The thalamus is a pacemaker of synchronous brain oscillations during certain (non-dreaming) phases of sleep, which depends on I_{Ca-T} underlying rhythmic bursting at a few hertz (Steriade and Llinas 1988; Steriade et al. 1993; Wang and Rinzal 1992; Wang et al. 1995; Destexhe and Sejnowski 2001). Thus, understanding the ion channel mechanism of postinhibitory bursting represents an example of how single-cell properties shape collective dynamics of the whole brain. Waking from sleep is correlated with depolarization of thalamic neurons, which in behaving conditions operate in the tonic mode, although rebound bursts may occur occasionally in wakefulness (Ramcharan et al. 2005).

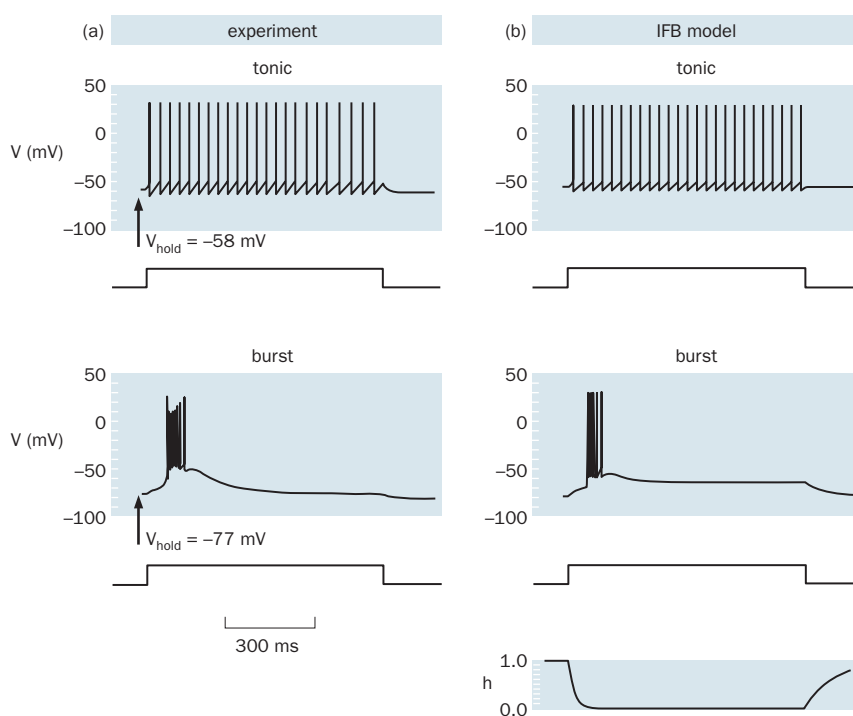


Figure 2.17 Bursting as postinhibitory rebound. Experimental records of relay neuron responses to depolarizing current steps and comparable integrate-and-fire-or-burst (IFB) model simulations. (a) A thalamic relay neuron responds in either tonic or burst mode depending on the initial membrane potential (V_{hold}). (b) The IFB model reproduces both tonic and burst responses depending on the initial membrane potential. Bottom: h , the deinactivation gating variable for I_{Ca-T} , during the IFB model burst response. Adapted with permission from Smith et al. (2000). [Smith, G. D., Cox, C. L., Sherman, S. M., & Rinzal, J. (2000). Fourier analysis of sinusoidally driven thalamocortical relay neurons and a minimal integrate-and-fire-or-burst model. *Journal of Neurophysiology*, 83(1), 588–610.]

Smith et al. (2000) proposed an extension of the LIF model to incorporate postinhibitory rebound. I_{Ca-T} is added to the voltage equation, a second equation describes the dynamics of its slow inactivation:

$$c_m \frac{dV}{dt} = -g_L(V - V_L) - g_{Ca-T} m_\infty(V) h(V - V_{Ca}) + I_{app} \quad (2.37)$$

$$\frac{dh}{dt} = \begin{cases} -h / \tau_h^- & \text{if } V > V_h \\ (1 - h) / \tau_h^+ & \text{if } V < V_h \end{cases} \quad (2.38)$$

where I_{Ca-T} has a reversal potential of $V_{Ca} = 120$ mV and activation threshold of $V_h = -60$ mV. For the sake of simplicity, this model assumes that the fast activation variable m is replaced by its steady-state (m_∞), which is zero for $V < V_h$ and 1 for $V > V_h$. The opposite holds true for the steady-state of the inactivation variable h . The time constant $\tau_h^+ = 100$ ms controls the timescale of hyperpolarization needed for postinhibitory rebound, and $\tau_h^- = 20$ ms determines the duration of depolarization underlying rebound burst. The model reproduced the salient tonic and burst modes of thalamic relay neurons (**Figure 2.17b**). This model has been used for other neuron types endowed with postinhibitory rebound, such as neurons in the subthalamic nucleus and external globus pallidus of basal ganglia (Wei et al. 2015).

2.5.3 Clustered and Irregular Spiking

Bursting denotes a brief period of spiking at a high frequency, a variant of which is clusters of spikes inter-nested in time with episodes of subthreshold membrane fluctuations (Rinzel 1987; Wang 1993). **Figure 2.18a** (upper panel) shows an example of such firing patterns of an inhibitory neuron in rat medial septum (Serafin et al. 1996). The regular repetition of spiking clusters is in the range of theta oscillations, a prominent rhythm in the hippocampus that occurs during an animal's spatial navigation and depends on inputs from the medial septum (Vanderwolf 1969; Buzsáki 2002). Interestingly, the medial septum contains two major cell types. Cholinergic cells release acetylcholine, a major neuromodulator, to neurons in the hippocampus, thereby regulating the network's excitability; on the other hand, GABAergic cells act on a faster timescale and entrain hippocampal neurons at theta oscillations (Stewart and Fox 1990). The intrinsic rhythm of spike clustering in GABAergic neurons of the medial septum thus represents a pacemaking function for the entire brain in a particular behavioral state (spatial navigation).

Such pacemaker activity of single neurons is captured by a Hodgkin-Huxley-type model endowed with a slowly inactivating potassium current (I_{KS}) (Wang 1993), as shown in **Figure 2.18a** (lower panel) (Wang 2002a). The mechanism relies on two assumptions. First, a low-threshold K^+ current I_{KS} should be transient and inactivates slowly during depolarization. Second, the neuronal spike afterhyperpolarization is strong, so that the average membrane potential is lower during spikes than during subthreshold epochs. Thus, when the neuron fires a cluster of spikes, the de-inactivation of I_{KS} slowly builds up during spike afterhyperpolarization, so the amplitude of I_{KS} increases gradually and eventually becomes large enough to terminate a spiking episode. When the cell does not fire spikes in a subsequent subthreshold epoch, the membrane potential is relatively depolarized, and I_{KS} slowly decreases due to inactivation, until the cell is sufficiently recovered and can start to fire again (**Figure 2.18b**). The subthreshold oscillations are produced by the interplay between a Na^+ current and the low-threshold activation of I_{KS} .

Spike clusters interspersed with subthreshold epochs can also take place irregularly rather than rhythmically, as observed in VIP/CR type of

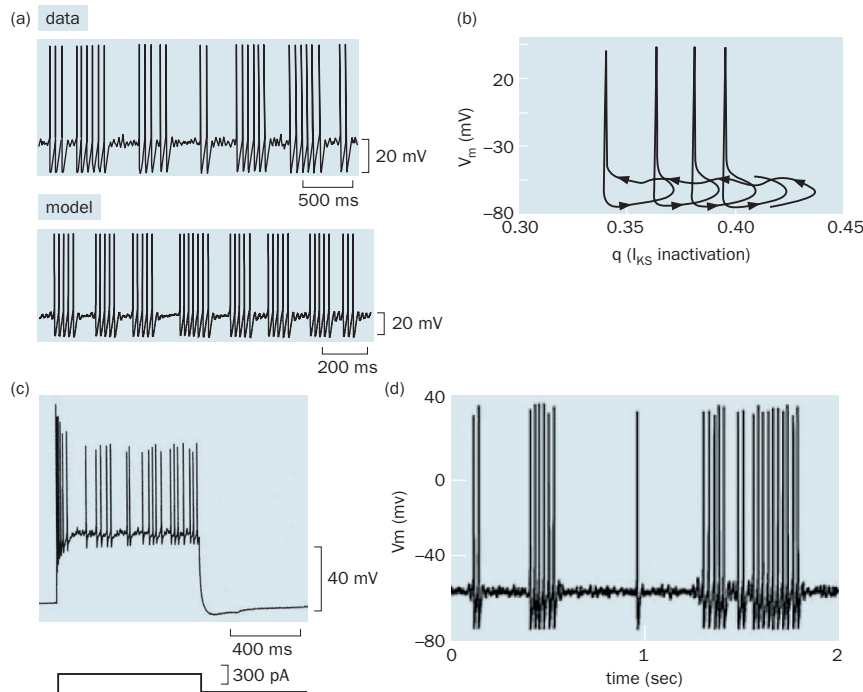


Figure 2.18 Spike clustering. (a)

Upper panel: A non-cholinergic (putative inhibitory) cell in the rat medial septum displays rhythmic alternations at theta frequency between “clusters” of spikes and epochs of subthreshold membrane potential oscillations. Lower panel: A Hodgkin-Huxley-type conductance-based model of the neuron shown in the upper panel. (b) For the model, membrane potential versus the inactivation gating variable q for a low-threshold potassium channel (I_{KS}). I_{KS} gradually de-inactivates (an increase of q , hence I_{KS}) during the hyperpolarizing phases of spikes (rightward arrows), whereas it inactivates (decreasing q) during subthreshold membrane oscillations (leftward arrows). (c) A VIP/CR interneuron in the neocortex shows spike clustering interspersed with subthreshold epochs. (d) A model of VIP/CR neuron with the Hodgkin-Huxley conductances and a low-threshold potassium current, driven by noisy inputs. The upper part of panel (a) is reproduced from Serafin, M., Williams, S., Khateb, A., Fort, P., & Mühlethaler, M. (1996). Rhythmic firing of medial septum non-cholinergic neurons. *Neuroscience*, 75(3), 671–675; the lower part of panel (a) and panel (b) are reproduced from X. J. (2002). Pacemaker neurons for the theta rhythm and their synchronization in the septohippocampal reciprocal loop. *Journal of Neurophysiology*, 87(2), 889; panel (c) is reproduced from Cauli, B., Audinat, E., Lambolez, B., Angulo, M. C., Ropert, N., Tsuzuki, K., ... & Rossier, J. (1997). Molecular and physiological diversity of cortical nonpyramidal cells. *Journal of Neuroscience*, 17(10), 3894–3906; panel (d) is reproduced from Stiefel, K. M., Englitz, B., & Sejnowski, T. J. (2013). Origin of intrinsic irregular firing in cortical interneurons. *Proceedings of the National Academy of Sciences of the United States of America*, 110(19), 7886–7891.

inhibitory neurons in the neocortex (Figure 2.18c) (Cauli et al. 1997; Markram et al. 2004). It is interesting that single isolated cells, in the absence of synaptic inputs, can display irregular firing dynamics, which may be deterministic chaos (Strogatz 2016) or a result of intrinsic noise (White et al. 2000). A conductance-based model with a low-threshold potassium current, driven by a noisy input current, produces irregular firing patterns similar to experimental observations (Figure 2.18d) (Stiefel et al. 2013).

Modern genomic analysis of the neural system has made it possible to link genes to physiological actions of their encoded proteins. An early example was provided from a study on the single cells of a central pattern generator of crustaceans, showing a strong positive correlation between mRNA abundance and the corresponding ion channel conductance measured by voltage-clamp experiments (Schulz et al. 2006). Figure 2.19a shows the result for three types of ion channels: non-inactivating potassium current I_{Kd} described in the Hodgkin-Huxley model, an inactivating potassium current I_A , a voltage- and calcium-activated potassium current I_{BK-KCa} . From ion channels one can deduce their physiological effects on a single cell. In particular, the width of an action potential is determined by I_{Kd} , its corresponding gene is called Kv3.1 or KCNC1 (Rudy 1988). The spike width is narrower in PV-expressing interneurons than in SST-expressing ones in the cortex, thus the gene that encodes I_{Kd} should have a higher expression in PV than in SST cells. That is indeed the case, as shown in Figure 2.19b. However, there are several intermediate steps from a gene to its encoded protein to physiology. It remains to be seen to what extent the expression level of a gene is a good predictor for the physiological effect of its encoded protein.

A remarkable advance in this area is to use single-cell transcriptomic data for cell-type classification in the mammalian cortex, which has been recently reported for mouse (Yao et al. 2023) and macaque monkey (Chen et al. 2023a). In the macaque cortex, data were collected using single-nucleus RNA-seq measurements, and a clustering method was applied to the database, revealing 264 transcriptomic clusters for cell-type taxonomy of the macaque cortex (Figure 2.19c). The dendrogram depicts 264 cell types (annotated in the outer circle) based on the median expression levels

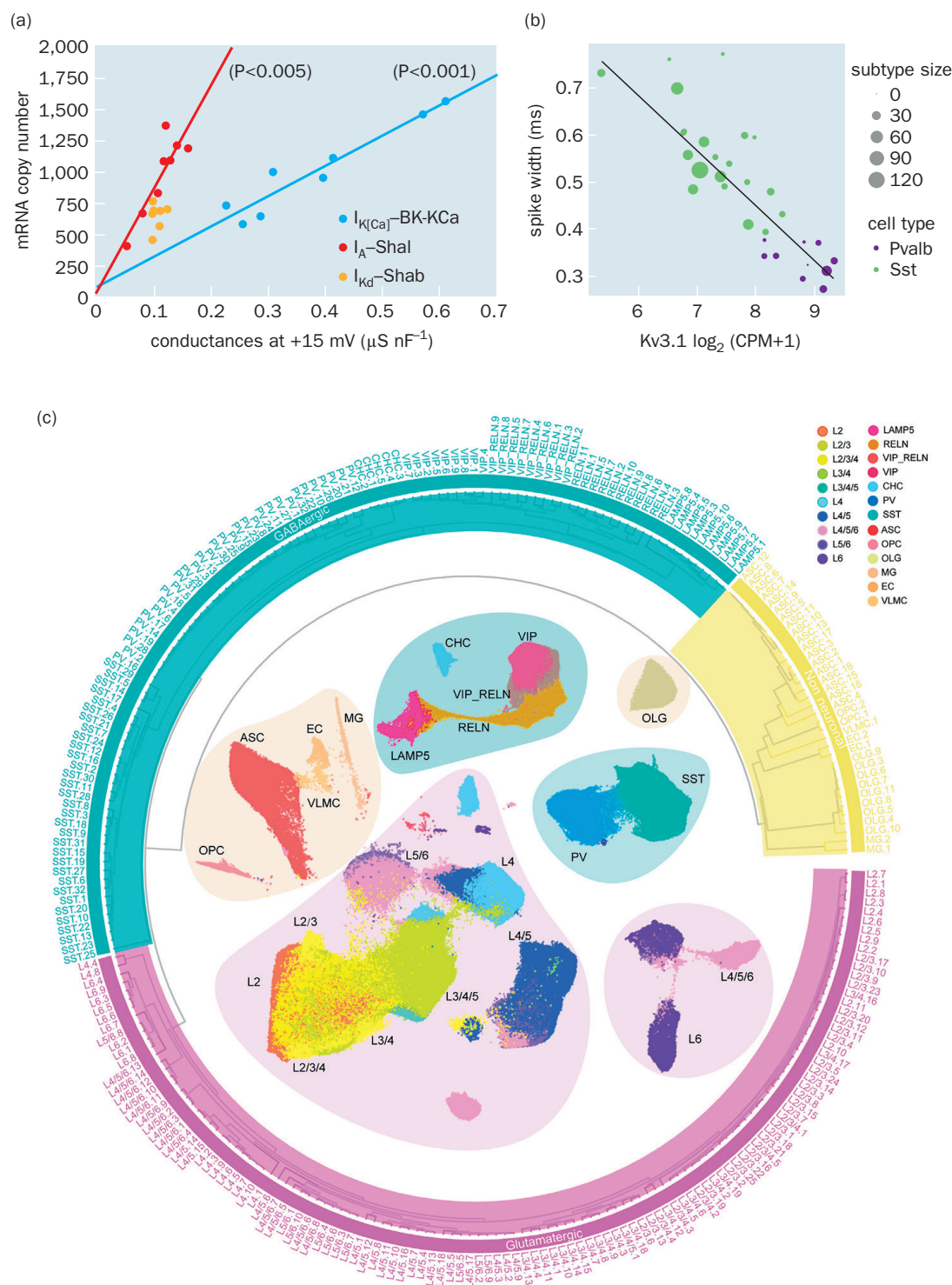


Figure 2.19 Genomic analysis for cell-type classification and single cell properties. (a) Ion channel mRNA abundance is correlated with measured membrane conductance in neurons of a crab's central pattern generator; each data point is for an individual cell. (b) The spike width (at half of the maximal voltage) is negatively correlated with the expression level of the gene Kv3.1 (KCNC1) that encodes the potassium conductance underlying spike repolarization of PV and SST positive GABAergic cells in the mouse V1. Each bubble represents a subcell type defined in Tasic et al. (2018); the size of each bubble represents the number of cells within that subcell type. (c) Transcriptomic cell-type taxonomy of 146 cortical regions in macaque brain. CHC, chandelier cells; ASC, astrocyte; MG, microglia; OLG, oligodendrocyte; OPC, oligodendrocyte precursor cells; EC, endothelial cells; VLMC, vascular leptomeningeal cell. Panel (a) is reproduced from Schulz et al. (2006), (b) is provided by Yijie Kang with data from Gouwens et al. (2020), (c) is reproduced from Chen et al. (2023a). [Schulz, David J., Jean-Marc Goaillard, & Eve Marder. Variable channel expression in identified single and electrically coupled neurons in different animals. *Nature Neuroscience* 9.3 (2006): 356–362.] [Chen, Ao, Yidi Sun, Ying Lei, Chao Li, Sha Liao, Juan Meng, Yiqin Bai et al. Single-cell spatial transcriptome reveals cell-type organization in the macaque cortex. *Cell* (2023), 186, <https://doi.org/10.1016/j.cell.2023.06.009>]

of top 50 marker genes, revealing the hierarchical clustering of three major classes (glutamatergic, GABAergic, and non-neuronal types; marked by background colors) and 23 cell subclasses (marked by dot colors). This database offers a wealth of information to be mined by the community. Future genomic data analysis will also help our studies of the neural circuits described in later chapters.

In summary, single neurons display a plethora of firing patterns, with potential functional implications such as spike frequency adaptation and rhythmogenesis. In the neocortex, SRA is typically weak in PV inhibitory neurons (which are hence often called “fast spiking” cells), but prominently displayed by excitatory pyramidal neurons (McCormick et al. 1985), with some in deep layers showing burst firing. SST/CB inhibitory neurons also show SRA (Kawaguchi and Kubota 1993; Gibson et al. 1999), whereas VIP/CR inhibitory neurons are characterized by irregular firing (Cauli et al. 1997; Mendonça et al. 2016). Thus, we have now become acquainted with models of major cell types of excitatory and inhibitory neurons in the neocortex. They provide neuronal units in a variety of network models. But before we discuss neural networks, it is necessary to learn how to mathematically describe synaptic interactions between nerve cells.

2.6 SINGLE SYNAPSE MODELS

A chain of physiological events takes place when a spike traveling along an axon arrives at a synapse (Katz 1966). During an action potential, the strong membrane depolarization of the axonal terminal activates voltage-gated calcium channels, resulting in a brief and large calcium influx into the presynaptic intracellular space (Katz and Miledi 1968; Llinás 1982). The latter triggers release of neurotransmitters as a stochastic process with some probability. The availability of transmitters at the presynaptic terminal and the release probability are activity dependent, which give rise to short-term synaptic plasticity (see Section 2.7). Released transmitters diffuse through the synaptic cleft and bind with receptors at the postsynaptic site, thereby imparting on the membrane of the targeted cell. In the central nervous system, at an excitatory synapse the primary transmitter is glutamate, and major types of glutamate receptors include AMPA (α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptor) and NMDA (N-Methyl-D-aspartic acid) receptors as well as mGluR (G-protein-coupled metabotropic glutamate) receptors. On the other hand, at an inhibitory synapse, the primary transmitter is GABA (γ -Aminobutyric acid), and major types of GABA receptors are GABA_A and GABA_B receptors (Nichols et al. 1992; Koch 1999). The binding of a transmitter with its receptor opens an ion channel on the postsynaptic membrane, resulting in either an excitation (a depolarization mediated by glutamate receptors) or an inhibition (a hyperpolarization mediated by GABA receptors). Therefore, neuronal communication requires a combination of electrical signals (spikes) and chemical substances, and it is inherently unidirectional (there is no principled reason that if one neuron sends signals to another neuron, the communication is reciprocated). Moreover, in the central nervous system, a given neuron only synthesizes either glutamate or GABA. Therefore, its action on all its connected postsynaptic neurons is uniformly depolarizing or hyperpolarizing. In other words, a neuron is either excitatory or inhibitory, not both. This is the “Dale’s law,” named after the British physiologist Henry Hallett Dale (Nichols et al. 1992).

Just like a diversity of models for single neurons, modeling single synapses may take into account a varying degree of complexity of the synaptic transmission process. Here, we will discuss mathematical descriptions of synaptic transmission in multiple ways. In those descriptions, the focus is

on the impact of a presynaptic spike onto a postsynaptic neuronal membrane. The time taken by all the intermediate steps of synaptic transmission is lumped into a single parameter called “transmission latency” Δ_L , which is small and neglected sometimes by assuming $\Delta_L = 0$.

Two major respects differentiate various models of single synapses. First, a distinction must be made between “current-based” and “conductance-based” synapse models. A conductance-based model describes a synaptic current similarly to any other ionic current, as $I_{syn} = g_{syn}s(V - V_{syn})$, where g_{syn} is the maximum conductance, and s is a synaptic gating variable that is not activated unless there is a presynaptic spike. Therefore, a synaptic current depends on both presynaptic activity and postsynaptic membrane V . The reversal potential determines whether a current is depolarizing or hyperpolarizing; typically, $V_E = 0$ mV for synaptic excitation mediated by the AMPA and NMDA receptors of the transmitter glutamate, and $V_I = -70$ mV and -80 mV for synaptic inhibition mediated by the $GABA_A$ and $GABA_B$ receptors, respectively, of the transmitter GABA.

For a neuron that receives synaptic inputs from M presynaptic neurons, each with a weight w_i and gating variable s_i (which depends on the firing of presynaptic neuron $i = 1, \dots, M$), the summated synaptic current is written as:

$$I_{syn} = \sum_{i=1}^M w_i s_i (V - V_{syn}), \quad (2.39)$$

A current-based model neglects the dependence on the postsynaptic voltage, so instead of Eq. (2.39), we have:

$$I_{syn} = \sum_{i=1}^M J_i s_i, \quad (2.40)$$

where synaptic weight is denoted as J_i instead of w_i with a different physical unit. A current-based model is simpler than a conductance-based model, but it is worth keeping in mind that it does not account for conductance changes due to synaptic inputs, which may vary considerably depending on the brain states and stimulation conditions (Destexhe et al. 2003).

Secondly, choosing how to model the synaptic gating variable s involves a tradeoff between the desire for mathematical simplicity and the need for certain details to capture aspects of synaptic dynamics. This will become clear when we present several types of synapse models in progressively more complicated forms.

2.6.1 Kick Synapses

We will start with the simplest description of synaptic interaction, by assuming that each presynaptic spike causes an instantaneous change (a “kick”) in the postsynaptic neuron. For a current-based kick synapse model, the impact is directly on the voltage. Consider a presynaptic neuron, which a train of spikes at time t_j , $j = 1, 2, 3, \dots$, then the synaptic current is $I_{syn} = Js$ and $s = \sum_j \delta(t - t_j)$. Note that since the physical unit of a delta function is $1/s$, the unit of J is electric charge (current has the unit of charge per unit time).

If an LIF neuron is initially at rest, and a presynaptic cell fires a single spike at time $t = 0$, then by integrating Eq. (2.1) one obtains:

$$V(t) = V_L + \frac{J}{C_m} \exp\left(-\frac{t}{\tau_m}\right) H(t), \quad (2.41)$$

where $H(t)$ is a Heaviside function, $H(t) = 0$ if $t < 0$ and 1 if $t > 0$. Thus, the postsynaptic potential (PSP) produced by each presynaptic spike consists of an instantaneous kick of size $\bar{J} = J / C_m$ followed by an exponential decay with time constant τ_m . For example, if the synapse is excitatory (J is positive) and the unitary charge is $J = 0.04$ pC, and $C_m = 0.2$ nF, then the kick size is $\bar{J} = 0.04 / 0.2 = 0.2$ mV.

With a spike train fired by a presynaptic neuron, the impacts are additive. Moreover, if M neurons synapse onto a postsynaptic cell, each with a weight J_i ($i = 1, \dots, M$), then the current into the postsynaptic neuron is additive over all the presynaptic neurons, $I_{syn} = \sum_{i=1}^M J_i s_i$ and $s_i = \sum_j \delta(t - t_j^i)$, where t_j^i is the time of the j^{th} spike from the i^{th} presynaptic neuron.

As a simple example, assume that all synaptic weights are the same ($J_i = J$) and spike trains are Poisson processes, with a rate λ_i for presynaptic neuron i ($i = 1, 2, \dots, M$). Because the sum of Poisson processes is itself a Poisson process, the spike train of overall synaptic input is still a Poisson process with a rate $\lambda = \sum_{i=1}^M \lambda_i$. When the firing rate λ is low (say 1 Hz), a typical time interval between two synaptic inputs is large (~ 1 s) compared to τ_m (~ 10 ms), in which case V decays back to V_L after each synaptic input. On the other hand, with a large firing rate (e.g., $M = 100$ and $\lambda_i = 5$ Hz, therefore $\lambda = 500$ Hz), intervals between synaptic inputs are small (2 ms), and V does not have time to decay to baseline between synaptic inputs. Instead, the neuron integrates synaptic inputs with a filter characterized by τ_m . With sufficiently strong excitation (determined by presynaptic activity and synaptic weight), a postsynaptic neuron can reach the firing threshold, generating an output spike train.

For a conductance-based synapse model, we have $I_{syn} = \sum_{i=1}^M w_i s_i (V - V_{syn})$. Inserting this input current into an LIF model (Eq. 2.1), it is clear that, as a major difference with a current-based synapse model, synaptic bombardments can significantly affect the neuronal time constant. Specifically, if synaptic inputs have an average conductance of G_{syn} , so the average $I_{syn} \sim G_{syn}(V - V_{syn})$, then the effective time constant is $\tau_{m,eff} = C_m / (G_L + G_{syn})$, which can be much shorter than $\tau_m = C_m / G_L$ without synaptic inputs (Destexhe et al. 2003). One of the implications is that with increased synaptic inputs, the neuron becomes leakier (with a larger effective leak conductance), thereby less capable of temporal integration. This implies that a neuron may not be able to integrate synaptic inputs over tens of milliseconds. Instead, its firing is determined by the coincidence of multiple synaptic inputs within a very brief period of time, in which case a neuron would behave as a “coincidence detector” rather than as an “integrator” (Abeles 1991; Softky 1994).

2.6.2 Filter and Kinetic Models of Synaptic Transmission

Kick synapse modeling does not account for the temporal properties of synaptic transmission, which can be measured with voltage-clamp technique. If the membrane potential is fixed and a presynaptic neuron is triggered to fire a spike or several spikes, one can measure $I_{syn} / (V - V_{syn})$ to deduce the time course of the gating variable $s(t)$ itself, which has an onset and a decay as well as temporal integration ability for a train of spikes (Katz 1966; Nichols et al. 1992).

A variety of models have been used to account for the temporal synaptic dynamics in addition to transmission delay. First, one can assume a temporal filter $h(t)$, for a given presynaptic train $t_j, j = 1, 2, 3, \dots$, the synaptic variable $s(t)$ is given by

$$s(t) = \sum_j h(t - t_j) \quad (2.42)$$

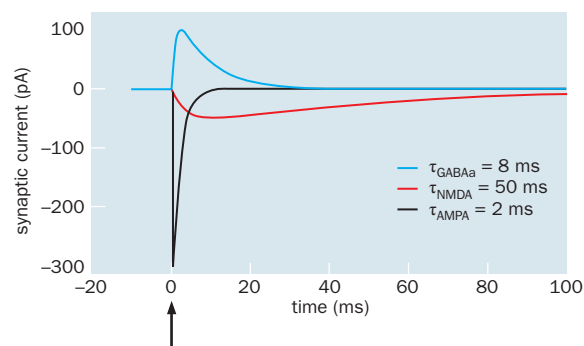
The kick synapse model is a special case, with $h(t - t_j) = \delta(t - t_j)$. A commonly used filter is $h(t) = (t / \alpha) \exp(-at)$. This “alpha-function” rises from zero, reaches a maximum at time $t_{peak} = 1 / \alpha$, then decays exponentially with a time constant $\tau_{syn} = 1 / \alpha$. If we want to describe the rise time and decay time with independent parameters, we can choose a difference of two exponentials, $h(t) = A(\exp(-t / \tau_1) - \exp(-t / \tau_2))$, where τ_1 is longer than τ_2 so that $h(t)$ is positive. The filter $h(t)$ rises from zero, peaks at $t_{peak} = \ln(\tau_1 / \tau_2) / (1 / \tau_2 - 1 / \tau_1)$ and afterwards decays as $\sim \exp(-t / \tau_1)$. **Figure 2.20** displays the time course of synaptic response to a single presynaptic spike for the AMPA receptor and NMDA receptor-mediated excitation, and GABA_A receptor-mediated inhibition according to this description of a difference of two exponentials.

Synapses can also be described by a kinetic equation, assuming that channels that bind with a receptor-like glutamate or GABA obey a similar first-order chemical reaction as those for intrinsic ion channels (Eq. 2.10), except that it is activated by a presynaptic neuron that fires spikes (Wang and Rinzel 1992; Destexhe et al. 1998). For a presynaptic spike train $t_j, j = 1, 2, 3, \dots$:

$$\frac{ds}{dt} = \alpha_s \sum_j \delta(t - t_j)(1 - s) - \beta_s s \quad (2.43)$$

where the opening rate $\alpha_s \sum_j \delta(t - t_j)$ is zero except when there is presynaptic spiking activity, and β is the rate of “unbinding” with the receptor leading to channel closing; in the absence of spiking, $s(t)$ decays exponentially with a time constant $\tau_s = 1 / \beta_s$. Note that when s is close to 1, the first term in Eq. (2.43) approaches 0, thus s cannot exceed 1, reflecting the fact that there is only a finite amount of receptor-channel complex available at a synapse. The saturation property distinguishes the kinetic model from the filter models in which the synaptic current can grow without bound. This formulation can be extended to a sequence of two or more chemical reactions so that $s(t)$ displays different onset and decay times as well as summation to saturation properties that can be fit with experimental measurements (Destexhe et al. 1998; Wang 1999b).

Figure 2.20 Model of an excitatory synaptic current mediated by the AMPA (black) or NMDA (red) receptor and an inhibitory synaptic current mediated by the GABA_A receptor (blue). With a clamped voltage, the time course of synaptic response to a single spike at $t = 0$ is determined by the kinetics of the gating variable s of synaptic conductance. An excitatory current is negative (inward), whereas an inhibitory current is positive (outward). Note different rise time and decay time course for each of the three currents.



To understand better synaptic saturation, let us approximate the pre-synaptic spike train by its average rate $r = (1/T) \int_0^T \sum_j \delta(t-t_j)$. Then Eq. (2.43) becomes:

$$ds/dt = \alpha_s r(1-s) - \beta_s s. \quad (2.44)$$

Thus with repetitive stimulation, s increases according to this linear equation and eventually reaches a steady-state:

$$s_{ss} = \frac{\alpha_s \tau_s r}{\alpha_s \tau_s r + 1} \quad (2.45)$$

The level of synaptic activation increases roughly linearly with the pre-synaptic firing rate as long as $\alpha_s \tau_s r \ll 1$ so that the denominator is ~ 1 . For the sake of simplicity, consider $\alpha_s \sim 1$, when this condition of linearity becomes $r \ll 1/\tau_s$. This holds true when the firing rate is tens of Hz, for the fast AMPA receptor-mediated excitation (with $\tau_{AMPA} \sim 2$ ms (Hestrin et al. 1990b; Spruston et al. 1995), $1/\tau_{AMPA} \sim 500$ Hz) or GABA_A receptor-mediated inhibition (with $\tau_{GABA} \sim 10$ ms (Salin and Prince 1996; Xiang et al. 1998), $1/\tau_{GABA} \sim 100$ Hz). This is why we often can neglect the factor $(1-s)$ in Eq. (2.43), so it becomes $ds/dt = \alpha_s \sum_j \delta(t-t_j) - \beta_s s$ for AMPA and GABA_A receptor-mediated synaptic currents, which is equivalent to a linear filter with the time constant τ_s . One advantage of the simplification is that if M presynaptic neurons converge on a postsynaptic cell, one does not need to write M equations for $s_i, i=1,2,\dots,M$. Instead, the summated inputs can be described by a single variable $s_{tot} = \sum_{i=1}^M s_i$, which obeys

$ds_{tot}/dt = \alpha_s \sum_i \sum_j \delta(t-t_j^{(i)}) - \beta_s s_{tot}$. The approximation $(1-s) \sim 1$ is no longer valid in the case of the NMDA receptor-mediated excitation with a time constant of 50–100 ms (Hestrin et al. 1990b; Spruston et al. 1995).

2.6.3 NMDA Receptor-Mediated Synaptic Excitation

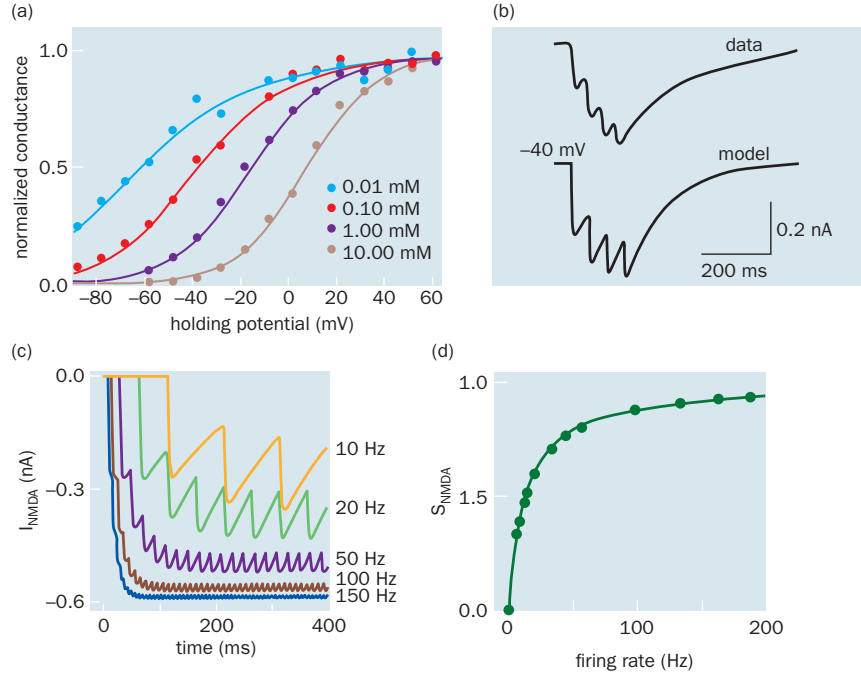
A distinguishing feature of the NMDA synaptic conductance is its nonlinear dependence on the postsynaptic voltage, so the current is written as:

$$I_{NMDA} = g_{NMDA} F(V) s(V - V_E) \quad (2.46)$$

where $F(V) = 1 / (1 + [\text{Mg}^{2+}] \exp(-0.062V) / 3.57)$ and $[\text{Mg}^{2+}]$ is the extracellular magnesium concentration (~ 1 mM in the central nervous system). **Figure 2.21a** shows this sigmoid-shaped dependence on the postsynaptic membrane potential, derived from single-channel experiments (Jahr and Stevens 1990). Thus, the NMDA channel opening is negligible if the membrane potential is close to resting state (~ -70 mV), and becomes large only with depolarization. In other words, the NMDA conductance activation requires a coincidence of presynaptic and postsynaptic activity. This property endows the NMDARs with a crucial role in Hebbian synaptic plasticity that relies on pre- and post-synaptic co-activation (see Chapter 4).

Furthermore, NMDA receptors also play a major role in online neural computation, especially in brain structures of importance to higher cognition (Wang 1999b). The NMDAR-mediated excitation is much slower than that mediated by the AMPA receptors, with $\tau_{NMDA} \sim 50 - 100$ ms (depending on the NMDA receptor subtypes). In this case, saturation cannot be neglected even for firing rates within the commonly observed physiological range (10–50 Hz). **Figure 2.21b–d** show the behavior of

Figure 2.21 NMDA receptor-mediated current. (a) Voltage dependence. (b) Time course of the current in response to a train of four spikes. (c) Current saturates at high input firing rate. (d) Average synaptic gating variable (steady-state) as a function of input firing rate. Panel (a) is reproduced from Jahr and Stevens (1990), upper trace of (b) from Hestrin et al. (1990a), lower trace of (b) and (c–d) from Wang (1999b). [Jahr, C. E., & Stevens, C. F. (1990). Voltage dependence of NMDA-activated macroscopic conductances predicted by single-channel kinetics. *Journal of Neuroscience*, 10(9), 3178–3182.] [Hestrin, S., Perkel, D. J., Sah, P., Manabe, T., Renner, P., & Nicoll, R. A. (1990, January). Physiological properties of excitatory synaptic transmission in the central nervous system. In *Cold Spring Harbor Symposia on Quantitative Biology* (Vol. 55, pp. 87–93). Cold Spring Harbor Laboratory Press.] [Wang, X. J. (1999). Synaptic basis of cortical persistent activity: The importance of NMDA receptors to working memory. *Journal of Neuroscience*, 19(21), 9587–9603.]



a model of NMDAR-mediated current with saturation. In **Figure 2.21b**, the model (lower panel) reproduces a sample trace of measured NMDAR-mediated current in a pyramidal neuron (upper panel). The voltage was clamped to -40 mV, and thus the observed current is negative and proportional to s . In response to four stimuli at 25 Hz, the increment of current by each stimulus gradually decreases, because s does not have time to decay back to zero between stimuli, and the amount of increase by each stimulus is proportional to $(1 - s)$. Biologically, the unbinding of channels from NMDARs is slow; subsequent transmitter release can only recruit those unbound (closed) channels. This effect is stronger with a higher stimulus rate (**Figure 2.21c**). The temporal average of s displays a firing rate dependence (**Figure 2.21d**), as predicted by Eq. (2.45). Notably, when the firing rate is higher than ~ 50 Hz, a further increase in presynaptic activity no longer recruits more excitatory drive. It is noteworthy that saturation does not occur for a single synaptic input (Mainen et al. 1999) but for a train of them at sufficiently high frequencies.

2.7 SHORT-TERM SYNAPTIC PLASTICITY

Short-term synaptic plasticity (Zucker 1989; Fisher et al. 1997; Abbott and Regehr 2004) refers to the phenomenon that synaptic transmission between two neurons is not static, but can either decrease (**Figure 2.22a**) or increase (**Figure 2.22b**) depending on the history of the presynaptic neuron's firing activity (Abbott and Regehr 2004).

Recall that synaptic transmission depends on transmitter release, but there is only a limited number of vesicles of transmitters at a presynaptic terminal. Consider a presynaptic terminal with a number $N \leq N_0$ of readily releasable vesicles of neurotransmitters. Upon the arrival of an action potential, a vesicle of neurotransmitters is released with certain probability P_{rel} , hence on average $P_{rel}N$ vesicles are released, and N is reduced to $N - P_{rel}N$. Between two stimuli, the pool of vesicles is gradually refilled, with a recovery time constant τ_D . Hence, the postsynaptic potential response (proportional to $P_{rel}N$) is initially $P_{rel}N_0$ but can exhibit short-term depression (**Figure 2.22a**) as N decreases with time, typically on a

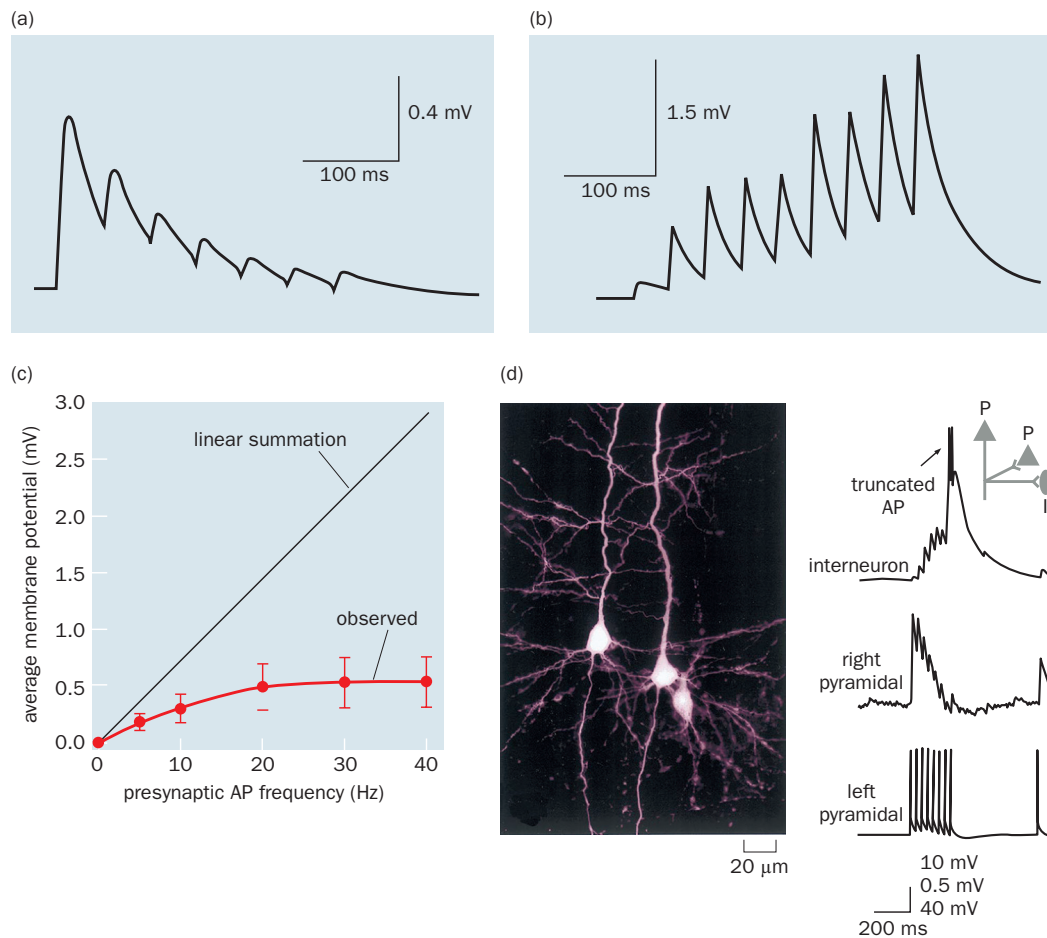


Figure 2.22 Short-term synaptic plasticity. (a) An example of short-term synaptic depression. Two connected neurons are recorded, showing the membrane potential of the postsynaptic neuron in response to a train of spikes in the presynaptic neuron. The amount of voltage increase is the largest at the first response and gradually decreases to a lower steady-state level. (b) An example of short-term synaptic facilitation. In this case, the voltage increase is the smallest at the first response and gradually grows to a higher steady-state level. (c) The average change of voltage as a function of the presynaptic firing rate. Unlike linear summation, with short-term depression measured neuronal response (red) reaches a plateau at a presynaptic firing rate above 20 Hz. (d) Triple recording from a presynaptic excitatory pyramidal neuron that projects to a postsynaptic pyramidal neuron and a postsynaptic inhibitory interneuron. Synaptic connection displays short-term depression for the former, facilitation for the latter, target cell. Panel (a) is reproduced from [Markram, H. & Tsodyks, M (1996) Redistribution of synaptic efficacy between neocortical pyramidal neurons. *Nature* 382(6594):807–810; panel (b) is reproduced from Markram, H, Wang, Y. & Tsodyks, MV (1998) Differential signaling via the same axon of neocortical pyramidal neurons. *Proceedings of the National Academy of Sciences of the United States of America*, 95(9):5323–5328. Copyright (1998) National Academy of Sciences, U.S.A.] [Tsodyks, M. V., & Markram, H. (1997). The neural code between neocortical pyramidal neurons depends on neurotransmitter release probability. *Proceedings of the National Academy of Sciences of the United States of America*, 94(2), 719–723. Copyright (1997) National Academy of Sciences, U.S.A.; panel (d) is reproduced from Markram, H, Wang, Y. & Tsodyks, MV (1998) Differential signaling via the same axon of neocortical pyramidal neurons. *Proceedings of the National Academy of Sciences of the United States of America*, 95(9):5323–5328. Copyright (1998) National Academy of Sciences, U.S.A.

timescale of hundreds of milliseconds. On the other hand, P_{rel} depends on the dynamical state of the transmitter release machinery, and can increase over time in response to a stimulus train, leading to short-term facilitation (Figure 2.22b).

Here we will discuss short-term synaptic depression and facilitation at the presynaptic site that occur on the timescale of hundreds of milliseconds to a few seconds. It is noteworthy that at the postsynaptic site, there exist dynamical changes such as receptor desensitization on comparable timescales (Hille 2001). Through these processes, synapses display short-term memory traces and can carry out interesting computations.

2.7.1 Short-Term Synaptic Depression

Synapses endowed with short-term depression are sensitive to changes in the input but not its steady-state. This can be seen using a simple phenomenological model (Dayan and Abbott 2001). Let $D = N / N_0$, then after each release in average D is reduced to $(1 - P_{rel})D$, and between release events D recovers towards one with a time constant τ_D . Mathematically, D obeys:

$$\frac{dD}{dt} = -P_{rel}D \sum_j \delta(t - t_j) + \frac{(1 - D)}{\tau_D} \quad (2.47)$$

where D multiplying $\delta(t - t_j)$ is understood to take the value D^- evaluated immediately before the spikes, at t_j^- . In this description, “during a spike,” which is a point in time, the δ function has an infinite amplitude, and the second term can be neglected. Thus, the update by a spike is given by $D^+ - D^- = -P_{rel}D^-$, or $D^+ = (1 - P_{rel})D^-$. Between spikes, $D(t)$ recovers exponentially towards 1 with a time constant τ_D .

For a Poisson spike train at rate r , if we approximate the presynaptic spike train $\sum_j \delta(t - t_j)$ by its rate r , the dynamical equation for D can be written as

$$\frac{dD}{dt} = -P_{rel}rD + \frac{1 - D}{\tau_D} \quad (2.48)$$

The steady-state of this equation is given by $dD / dt = 0$, which yields:

$$D_{ss} = \frac{1}{1 + P_{rel}r\tau_D}.$$

According to this result, for the input rate r higher than $1 / P_{rel}\tau_D$, the term $P_{rel}r\tau_D$ is larger than 1 and dominates the denominator. Therefore, $D_{ss} \simeq 1 / P_{rel}r\tau_D$, inversely proportional to r . The transmission rate (release per unit time) is the amount of release per spike, $P_{rel}D$, multiplied by the presynaptic firing rate r . Its steady state, hence the postsynaptic response, is independent of input rate r when $r \gg 1 / P_{rel}\tau_D$. For instance, if $P_{rel} = 0.6$ and $\tau_D = 500$ ms, then $1 / P_{rel}\tau_D = 1 / 0.3 \simeq 3$ Hz. These model predictions have been confirmed by experimental observations on synapses between neocortical neurons (**Figure 2.22c**) (Abbott et al. 1997; Tsodyks and Markram 1997).

Functionally, this means that the impact on the postsynaptic cell is not sensitive to the steady-state level of presynaptic activity. On the other hand, when there is a sudden change in the input, the synapse can detect it by a transient response before the slow depression has the time to react. This is illustrated in **Figure 2.23**, where the postsynaptic response (proportional to $P_{rel}rD$) is plotted as the presynaptic firing rate undergoes a series of step changes. Note first that regardless of the presynaptic firing rate 25, 100, 10, 40 Hz, the steady-state $P_{rel}rD$ is quite similar, as predicted before. Second, when there is a sudden jump in the presynaptic rate, because the depression process takes time to react, the synapse signals input change by a rigorous but transient response, thereby performing a time derivative of the input. When r is decreased or increased to $r + \Delta r$, D briefly remains $\sim 1 / r$ (where r is the rate prior to the jump), thus the response jump is $\sim \Delta r / r$, reflecting the change in the input relative to the baseline, rather than the absolute amount of input change. This feature is apparent in **Figure 2.23**, where the transient response is similar for the transitions from 25 Hz to 100 Hz and from 10 Hz to 40 Hz. In the first step change, $r = 25$, $\Delta r = 75$; in the second step change, $r = 10$ Hz and $\Delta r = 30$ Hz. In both cases, $\Delta r / r = 3$. The transient postsynaptic response is roughly the same during

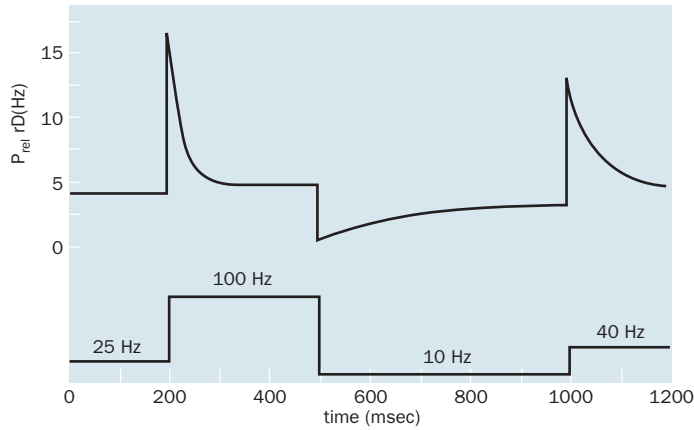


Figure 2.23 Synapse exhibiting short-term depression performs time derivative-like computation. The average postsynaptic response from a synapse model with short-term depression remains roughly the same for different long fixed presynaptic firing rate (r at 10, 25 or 100 Hz). It responds to stepwise change Δr with an amount proportional to the relative change of r , $\Delta r / r$. The parameters of the model are $P_{rel} = 0.6$ and $\tau_D = 500$ ms. Reproduced from Dayan, P. and L. F. Abbott (2001). *Theoretical Neuroscience*. MIT Press, Cambridge MA.

these two jumps of the presynaptic firing rate (Figure 2.23). Moreover, the time constant governing the decay process toward a steady-state is given by $\tau_D / (1 + P_{rel} r \tau_D)$, which is shorter at a higher input rate r . This explains why the transient response is longer lasting for the 10 to 40 Hz transition than for the 25 to 100 Hz transition. Therefore, short-term depression provides a synaptic mechanism for postsynaptic neurons to be responsive to changes in the input but not sensitive to constant input level, and contribute to adaptation in the visual system (Abbott et al. 1997).

2.7.2 Short-Term Synaptic Facilitation

Experiments showed that a presynaptic pyramidal cell connects to another pyramidal cell with short-term depression or an inhibitory neuron with short-term facilitation (Figure 2.22d) (Markram et al. 1998). This and other findings (Caputi et al. 2009) suggest the following rule: whether an excitatory synaptic connection displays predominantly short-term depression or facilitation is determined by the postsynaptic (excitatory or inhibitory) cell types. By contrast, short-term dynamics of an inhibitory connection depends on the presynaptic cell subclass, e.g., depression for PV+ cell-mediated inhibition and facilitation for SST+ cell-mediated inhibition (Silberberg and Markram 2007; Ma et al. 2012; Feldmeyer et al. 2018; Campagnola et al. 2022).

The release probability of transmitters P_{rel} depends on the intracellular calcium concentration at a presynaptic terminal. The latter may accumulate over repetitive stimulation due to slow residual calcium (Zucker 1989; Fisher et al. 1997), and therefore P_{rel} would increase in time, leading to gradually larger synaptic response (which on average is proportional to P_{rel}). Various models have been proposed for short-term facilitation (Bertram et al. 1996; Markram et al. 1998; Tsodyks et al. 1998; Matveev and Wang 2000). Here we will describe a phenomenological model following Tsodyks et al. (1998). Assume that the release probability $P_{rel} = u(t)$ is activity-dependent and obeys the following dynamical equation:

$$\frac{du}{dt} = U \sum_j \delta(t - t_j) (1 - u) + (U - u) / \tau_F \quad (2.49)$$

where U (less than one) is the baseline value of u or release probability. Again, $U(1 - u)$ multiplying the spike train $\delta(t - t_j)$ takes the value $U(1 - u^-)$ evaluated immediately before the delta functions, at t_j^- . Thus, the update from before to after a spike is given by $u^+ - u^- = U(1 - u^-)$, or $u^+ = u^- + U(1 - u^-)$. In the absence of presynaptic firing, the first term of Eq. (2.49) is zero, and u decays exponentially towards U with a time constant

τ_F . If τ_F is long relative to time intervals between presynaptic spikes, spike-triggered calcium can accumulate at the presynaptic terminal, and an increase of u over time leads to synaptic facilitation.

To see quantitatively how u depends on the presynaptic activity, let us replace the spike train $\sum_j \delta(t - t_j)$ by its firing rate r . Then, solving the steady-state equation yields:

$$u_{ss} \simeq \frac{U(1 + r\tau_F)}{1 + Ur\tau_F} \quad (2.50)$$

Thus, at low firing rates ($r \leq 1/U\tau_F$), the second term in the denominator is small compared to 1, and u_{ss} increases linearly with r . For firing rates much larger than $1/U\tau_F$, u_{ss} approaches 1.

A synapse can display both short-term depression and facilitation. In that case, the coupled equations for u and D are (Mongillo et al. 2008):

$$\frac{du}{dt} = U \sum_j \delta(t - t_j)(1 - u) + \frac{(U - u)}{\tau_F} \quad (2.51)$$

$$\frac{dD}{dt} = -uD \sum_j \delta(t - t_j) + \frac{(1 - D)}{\tau_D} \quad (2.52)$$

In the case, the change of synaptic variable s induced by a presynaptic spike (the parameter α_s in Eq. 2.43) is proportional to uD , or more precisely u^+D^- because of the subtlety in differentiating a variable before and after a spike (as a delta-function of time). Recall the sequential biological processes of synaptic transmission: after each spike-triggered calcium influx, the dynamical variable u is increased from u^- to u^+ , which should be used as P_{rel} for transmitter release. The amount of transmitter released is given by D^- , and therefore the size of the postsynaptic potential is evaluated using u^+D^- . After the transmitter release, D is updated from D^- to D^+ (using $P_{rel} = u^+$) as a result of vesicle depletion. To reiterate, by a spike input $u^+ = u^- + U(1 - u^-)$, $D^+ = (1 - u^+)D^-$ and the synaptic variable s is updated using $\alpha_s = \alpha_0 u^+D^-$ in Eq. (2.43) with a constant α_0 .

When the spike train is replaced by a time-varying firing rate r , the approximate equations become:

$$\frac{du}{dt} = Ur(1 - u) + \frac{(U - u)}{\tau_F} \quad (2.53)$$

$$\frac{dD}{dt} = -uDr + \frac{(1 - D)}{\tau_D} \quad (2.54)$$

$$\frac{ds}{dt} = \alpha_0 uDr(1 - s) - \frac{s}{\tau_s} \quad (2.55)$$

which are used in firing rate models endowed with short-term synaptic plasticity.

Figure 2.24 displays an example of synaptic response to a stochastic spike train that includes single spikes and brief bursts of spikes. Facilitation occurs maximally when spikes occur at a high rate within a brief burst in time. By contrast, single spikes do not facilitate synapses, and thus have a low probability of being transmitted to the postsynaptic side. In other words, a facilitating synapse can act as a detector of bursts while “ignoring” temporally isolated spikes among a mixture of single spikes and bursts (Lisman 1997; Wang 1999a; Matveev and Wang 2000).

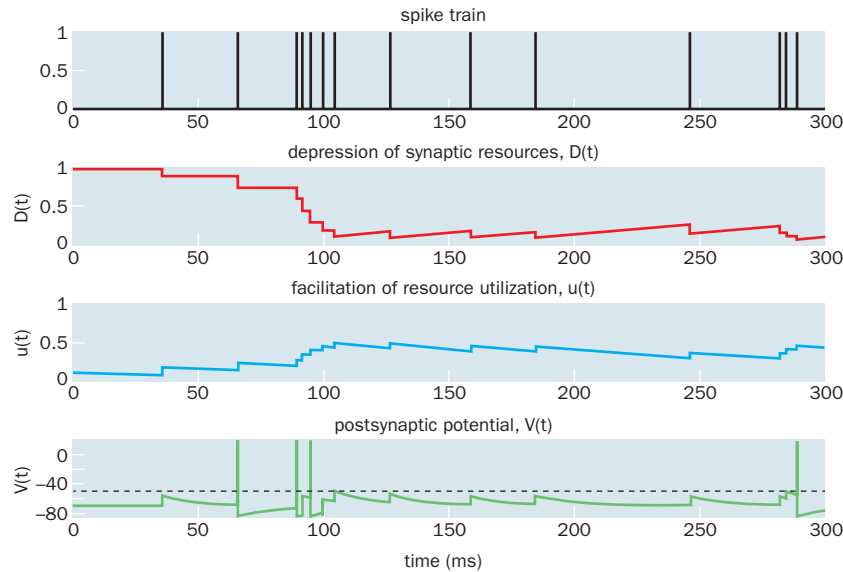


Figure 2.24 A model with a presynaptic neuron (top) and a postsynaptic neuron (bottom) connected by a synapse endowed with both short-term facilitation and depression. The depression variable $D(t)$ decreases while the facilitation variable $u(t)$ increases at each presynaptic spike. The combined effects result in a maximal synaptic drive and high probability of spiking in the postsynaptic neuron during a burst of spikes in the presynaptic neuron. $U = 0.1$, $\tau_F = 150$ ms and $\tau_D = 300$ ms. Model simulation by Victor Matveev based on Matveev and Wang (2000).

2.8 SUMMARY

In this chapter we introduced several models of neurons and synapses. We learnt some basic neurobiological concepts, such as membrane potential dynamics, ion channels, neurotransmitters and Dale's law. Single cells may exhibit various firing patterns such as adaptation and bursting. It is worth keeping in mind that single neuron properties vary from one brain region to another, as well as from one species to another, exemplified by unique features of pyramidal cells in the human cortex (Gidon et al. 2020; Galakhova et al. 2022). Synapses have their own temporal dynamics as well as short-term plasticity, which, if prominent, can accomplish interesting computation at the synaptic level, such as time-derivative of inputs or burst detection. We also covered mathematical concepts: time constant of a linear dynamical system, bifurcation in a nonlinear dynamical system, stability and attractors. These basic biological properties of neurons and synapses will serve as the building blocks of the brain. We will use them in models designed to investigate the complex dynamics of neural circuits and cognitive functions later in the book.

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